

IMPLEMENTATION PLAN · RESEARCHED BUILD GUIDE

# AMS

Step-by-Step Implementation Plan

Airline Management System — Finance · Fleet · Procurement  
Every step, every command, every verification gate

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| DOCUMENT           | Step-by-Step Implementation Plan                                       |
| COMPANION          | AMS-Specification.pdf — scope, glossary, architecture, frontier design |
| VERSION            | 1.0 — 30 September 2026                                                |
| PROJECT            | AMS — Airline Management System, multi-tenant SaaS                     |
| LANGUAGE           | JavaScript (ESM) · Express · Knex · PostgreSQL 16 · React              |
| STRUCTURE          | 7 waves · 20 migrations · 20 modules · 16 departments · 76 tables      |
| CURRENT STATE      | Wave 1 shipped — <b>118 tests passing, 0 type errors</b>               |
| RESEARCH BASE      | 14 primary sources, cited per step                                     |
| RESEARCHED CHANGES | <b>3 design corrections found during research — see Part II.3</b>      |

Confidential — Internal Project Documentation  
Every command in this document is executable as written. Every step states its verification.

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PART I

# How to Use This Plan

## 1.1 Relationship to the specification

This document is the executable companion to [AMS-Specification.pdf](#). The specification answers *what* the system is and why. This plan answers *exactly how to build it, in what order, and how to know each step worked*.

| Question                                                    | Answered by                                                        |
|-------------------------------------------------------------|--------------------------------------------------------------------|
| What is the domain vocabulary?                              | Specification Part III (glossary) and Part IV (abbreviation index) |
| What problem does each module solve, with an example?       | Specification Part II                                              |
| What are the advanced capabilities and how ready are they?  | Specification Part VII                                             |
| What is the architecture and why this stack?                | Specification Parts XIII–XIV                                       |
| Why does the domain behave this way? (regulatory grounding) | This plan, Part II                                                 |
| In what order do I write it, and how do I verify each step? | This plan, Parts IV–X                                              |
| What are the sources behind each decision?                  | This plan, Part XV                                                 |

## 1.2 Step conventions

| Marker   | Meaning                                                                                            |
|----------|----------------------------------------------------------------------------------------------------|
| SHIPPED  | Step is complete, tested and merged. Re-running it is a regression check, not a task.              |
| CRITICAL | Step is on the critical path. If it slips or is short-changed, downstream waves cannot be trusted. |
| RESEARCH | Step depends on an external source, supplier or regulator. Contains the citation.                  |
| TODO     | Not yet started.                                                                                   |

Every step follows the same five-part shape:

- 1. **Action** — what to build, in one sentence.
- 2. **Implementation** — the exact command, file, or SQL to execute.
- 3. **Output** — what exists when the step is done.
- 4. **Verification** — the command or assertion that proves it worked. A step with no verification is not done.
- 5. **Research basis** — where the requirement comes from, where relevant.

## 1.3 Effort units

| Unit | Meaning            | Typical tasks                                         |
|------|--------------------|-------------------------------------------------------|
| XS   | Under 2 hours      | Add a Zod field, add a test case, add an index        |
| S    | Half a day         | One endpoint plus tests, one migration, one component |
| M    | One to two days    | One module end to end with tests and UI               |
| L    | Three to five days | A subsystem requiring design review                   |

## 1.4 Prerequisites

```
# Verify toolchain before starting
node --version           # must be >= 20
npm --version
docker --version         # required for PostgreSQL, Redis, MinIO
docker compose version

# Clone and confirm the shipped baseline is green
cd ams
npm install
npm test                 # expect: 118 passed
npm run typecheck        # expect: 0 errors
```

## 1.5 What is already shipped

**WAVE 0 COMPLETE**

Wave 0 is the highest-risk wave because everything downstream depends on it being correct. It is done. These modules are built, tested and are the reason the remaining waves are a sequencing problem rather than a design problem:

- `shared/src/money.js` — integer-cent arithmetic, largest-remainder allocation
- `shared/src/metrics.js` — ASK, RPK, CASK, RASK, load factor, break-even
- `shared/src/costing.js` — flight cost, fleet-fixed allocation, route verdict
- `shared/src/airworthiness.js` — LLP controlling limit, AD/SB, MEL, check planning
- `shared/src/fuel.js` — burn reconciliation, reserves, tankering economics
- `shared/src/carbon.js` — CO<sub>2</sub>, CORSIA, EU ETS, the overlap deduction
- `backend/src/audit/chain.js` — hash-chained tamper-evident audit log
- `backend/src/modules/finance/cost-authority.js` — the control core
- Migrations 001–002 — tenancy, RLS, audit triggers, budgets, generated availability

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| W0 | W1 | W2 | W3 | W4 | W5 | W6 |
|----|----|----|----|----|----|----|

PART II

Research Base

Fourteen researched areas. Each states the primary source, what the source actually requires, and the constraint it places on the build. Steps in Parts IV–X cite these as **R1 – R14**.

R1 — Fuel planning components are prescribed, not discretionary

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UK CAA<br>SPO.OP.131<br>SPA.NCO.131 | <b>Source:</b> UK Civil Aviation Authority regulatory library, <i>Fuel and oil supply</i> . Corroborated by EASA fuel-planning guidance and the ICAO <i>Flight Planning and Fuel Management</i> seminar material.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Requires                            | Usable fuel for a flight is the sum of specific components: <b>taxi fuel</b> (expected use before take-off); <b>trip fuel</b> (take-off to landing); <b>contingency fuel</b> ; <b>destination alternate fuel</b> (destination aerodrome to alternate, or holding at destination where no alternate is required); <b>final reserve fuel</b> (for turbine-engined aeroplanes, fuel to fly <b>30 minutes</b> at holding speed 1,500 ft above the alternate, ISA conditions); <b>additional fuel</b> for ETOPS (FE ERA critical scenario, whichever of engine failure or loss of pressurisation requires the greater amount, at the most critical point on the route); <b>extra fuel</b> for anticipated delays or operational constraints; and <b>discretionary fuel</b> at the pilot-in-command's discretion. |
| Constraint                          | The fuel plan is <b>nine named components</b> , not three. AMS's <b>FuelPlan</b> must model all of them as discrete fields, because each is subject to different rules and different regulators. The 30-minute final reserve is a hard constant for turbine aeroplanes. <i>See steps 0.8, 3.6, 3.7.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

R2 — MEL rectification intervals are lettered categories with fixed day counts

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | EASA <b>CS-GEN-MMEL Issue 2</b> (Certification Specification for a Master Minimum Equipment List), Annex II to ED Decision 2020/012/R. Corroborated by the UK CAA MEL Compliance Document (Edition 8, February 2024).                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Requires   | MEL items carry a rectification interval designated by letter: <b>Category A</b> — no standard interval, any period specified other than calendar days, starting at the moment the defect is deferred; <b>Category B</b> — three calendar days; <b>Category C</b> — ten calendar days; <b>Category D</b> — one hundred and twenty calendar days. For B, C and D the interval <b>excludes the day of discovery</b> . The MMEL item table has five columns: system and sequence number, rectification interval, number installed, number required for dispatch, and remarks/exceptions — where remarks carry the <b>(M)</b> modification and <b>(O)</b> operating-limitation symbols. |
| Constraint | The "day of discovery is excluded" rule is an off-by-one that a naive <b>now + N days</b> calculation gets wrong on every CAT B item. Deferrals must be computed with an explicit exclusion of the discovery date. The (M)/(O) symbols are not commentary — an (O) item imposes an operating limitation that changes what the crew may do. <i>See steps 3.10, 3.11.</i>                                                                                                                                                                                                                                                                                                             |

THE SINGLE MOST IMPORTANT SENTENCE IN THE SOURCE MATERIAL

"The MEL cannot deviate from Airworthiness Directives or any other additional mandatory requirements." — EASA CS-GEN-MMEL

This means a MEL deferral can **never** be used to cure an outstanding Airworthiness Directive. The two are not merely different in kind — one is explicitly subordinate to the other. AMS must therefore never offer an operator the option of deferring an AD through the MEL path, and must not let a MEL deferral clear an AD compliance item. This is a design constraint discovered in research and it is implemented in step 3.9.

R3 — Airworthiness Directive structure is more than "due date"

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | US <b>14 CFR Part 39</b> as published in the Federal Register (79 FR 19848, April 2014; 87 FR 67354, November 2022); EASA <b>AD 2026-0125-E</b> (Emergency), <b>AD 2023-0167R1</b> , <b>AD 2024-0213</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Requires   | A real AD carries far more structure than a due date. Observed across the sampled documents: <b>Applicability</b> — specific aircraft models, specific serial numbers, and specific production modifications embodied (e.g. "all serials on which modification 21202 has been embodied, except those on which modification 152569 has been embodied"); <b>Reason</b> — the safety finding that triggered it; <b>Compliance time</b> — expressed as absolute thresholds, e.g. "before exceeding 24,000 flight cycles or 48,000 flight hours, whichever occurs first since the airplane's first flight"; <b>Repetitive inspection interval</b> — e.g. every 7,500 FC or 15,000 FH, whichever occurs first; <b>Corrective action</b> ; <b>ATA chapter code</b> (53 Fuselage, 34 Navigation, 32 Landing Gear); <b>referenced SB</b> as the accomplishment instruction; <b>non-cumulative tolerance</b> (5%) to allow synchronisation with other maintenance; <b>terminating action</b> — where a component manufacturer repair ends the repetitive inspections; and <b>supersession</b> where a later AD replaces it. |
| Constraint | An AD model with only "reference, type, due date" is inadequate and dangerous. AMS needs: modification-based applicability, multiple compliance thresholds, a repetitive interval, ATA chapter, non-cumulative tolerance, terminating action, and supersession. Emergency ADs additionally impose an <i>operating limitation</i> pending the first inspection, which is a dispatch limitation rather than a maintenance task. <i>See steps 3.8, 3.9, 3.12.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

R4 — ICAO Annex 6 is the densest single source of required behaviour

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | ICAO <b>Annex 6 — Operation of Aircraft</b> , Parts I–IV, 11th edition and amendments (icao.int). Structure: Part I International Commercial Air Transport — Aeroplanes; Part II International General Aviation — Aeroplanes; Part III International Operations — Helicopters; Part IV RPAS (adopted April 2024).                                                                                                                                                                   |
| Requires   | Operator certification and the Operations Manual (Attachment G); the Air Operator Certificate template and its Operations Specifications (Appendix 6, Attachment D); the Minimum Equipment List (Attachment E); authorisations (Attachment D-3); EDTO requirements (Attachment C); altimetry performance in RVSM airspace (Appendix 4); runway performance (Appendix 3); flight recorder data retrieval (newer amendments); Required Technical Safety Evaluations (Attachment D-1). |
| Constraint | AMS stores the AOC number, validity window and certification body on the tenant, and treats fleet authorisation as an operating constraint. The Operations Manual becomes a referenced document identifier on every procedure-bearing record, because an inspector asks "which manual section governs this?" and the answer must be one lookup, not a search. <i>See steps 0.9, 1.6, 3.12.</i>                                                                                      |

R5 — Personnel licensing is licence-type and validity specific

|            |                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | ICAO <b>Annex 1 — Personnel Licensing</b> ; <b>CAAZ</b> (Civil Aviation Authority of Zimbabwe) personnel licensing function — licenses aviation personnel and approves training organisations in accordance with ICAO and national requirements, working alongside AFCAC, the Department of Immigration, ZIMRA-Customs and the Meteorological Services Department.                |
| Requires   | Licences are type-specific (ATPL, MPL, ATPL(A), cabin crew) with associated type ratings, medical currency, and recurrent training currency. Regional bodies such as AFCAC add audit and harmonisation obligations on top of ICAO.                                                                                                                                                |
| Constraint | Expiry dates must be modelled as <b>three separate</b> fields on the crew record — licence, medical, recurrent training — because they expire on different cycles and a single "valid until" would let an aircraft be crewed with a lapsed medical. Licensing is also regional in practice, so the certification body is an attribute, not a constant. <i>See steps 1.7, 1.8.</i> |

R6 — Accountable managers are a legal requirement, not an org chart choice

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | South African Civil Aviation Regulations <b>127.06.2</b> (Commercial Helicopter Operations), which enumerates the appointed personnel a certificated operator must have approved by the Director: Chief Executive Officer; person responsible for flight operations; person responsible for aircraft; Air Safety Officer; Quality Assurance Manager; Security Manager. Corroborated by the Mauritius Department of Civil Aviation <i>Guidance for obtaining an Air Operator Certificate</i> (Issue 1 Rev 0), aligned to EU 2018/1139 and ICAO Doc 8335. |
| Requires   | These posts must be <i>appointed and approved</i> , not merely filled. Approval status is recorded. The applicant must additionally demonstrate adequate equipment, facilities and personnel, and that it can conduct operations safely and in compliance with all applicable rules.                                                                                                                                                                                                                                                                    |
| Constraint | This is the reason AMS has a 16-department registry with accountable-manager appointment records carrying regulatory approval state. It is not a design preference — an operator that cannot evidence its accountable managers cannot hold or renew its AOC. <i>See steps 1.5, 1.6, 1.9.</i>                                                                                                                                                                                                                                                            |

## R7 — Airline cost structure follows a documented industry taxonomy

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | US <b>BTS Table 4</b> , Quarterly US Scheduled Passenger Airlines Revenue, Expenses and Profits — 25 airlines, industry-standard expense proportions; IATA cost taxonomy and revenue accounting standards; <b>IATA Revenue Accounting Manual B16; ATA Spec 8</b> (ATA of America).                                                                                                                                                                                                                                                                       |
| Requires   | The observed US industry expense mix (3Q 2016, 25 carriers): <b>Labor 34.31%, Fuel 16.41%, Rentals 5.99%, Depreciation &amp; Amortisation 5.70%, Landing Fees 2.10%, Maintenance Materials 1.64%</b> , Transport-Related 12.42%, Other 20.35%. On the revenue side: passenger fares 75.05%, cargo 1.50%, baggage 2.48%, reservation changes 1.65%, transport-related 16.32%, other 2.99%.                                                                                                                                                                |
| Constraint | Two facts drive the design. First, <b>fuel is not the largest cost in this sample</b> — labour is, at roughly twice fuel. That is one reason AMS models crew cost as a first-class flight-attributable category rather than a period overhead, and it is a correction to the assumption most people carry into airline finance. Second, the categories are <b>industry-standard</b> , which is what makes any AMS CASK figure comparable to a third party. Any category AMS invents would be excluded from benchmarking. <i>See steps 0.2, 2.4, 5.8.</i> |

### WHY THIS CORRECTS AN ASSUMPTION IN THE SPECIFICATION

The specification states fuel is "typically 25–35% of operating cost". That figure is common in aviation commentary and is defensible for some carriers. It is not what the BTS industry aggregate shows, where labour exceeds fuel across the sector. AMS does not depend on which is true — it computes both from primary records per carrier — but the *design* must treat crew cost as flight-attributable, and it does.

## R8 — Traffic metrics have exact definitions that must be implemented precisely

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | IATA, <i>Understanding air traffic metrics</i> — Monthly Air Traffic Statistics covering 93% of world scheduled traffic; MIT/PMC paper <i>Operating costs and capacity in the airline industry</i> (activity-based costing for aviation).                                                                                                                                                                                                                                                                                                |
| Requires   | Definitions, stated exactly: <b>ASK</b> — seats available × distance flown (the supply side, unaffected by sales); <b>RPK</b> — paying passengers × distance (the demand side, and the only passenger-capacity measure carrying revenue); <b>PLF</b> = RPK ÷ ASK; <b>CTK</b> = tonnage × distance; <b>ACTK</b> = available cargo capacity × distance; <b>CLF</b> = CTK ÷ ACTK; <b>ATK</b> = combined passenger and cargo tonne-km; <b>FRT</b> = freight tonnes. Standard passenger weight applies when converting passengers to payload. |
| Constraint | Because ASK is <i>available</i> capacity and RPK is <i>revenue</i> demand, a revenue-per-ASK computed against boarded passengers rather than offered capacity is wrong. AMS must carry <b>sold</b> , <b>boarded</b> and <b>no-show</b> counts separately on <code>passenger_counts</code> , because PLF uses boarded passengers against offered seats, while revenue uses sold. Conflating them inflates or suppresses yield. <i>See steps 0.3, 2.5, 5.9.</i>                                                                            |

## R9 — Cost allocation in aviation is activity-based, with defined drivers

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | MIT / PubMed Central study of airline operating costs using activity-based costing; observed separation of costs into direct variable, direct fixed, and period costs.                                                                                                                                                                                                                                                                                                                                         |
| Requires   | The study separates costs by whether they are: attributable to a specific aircraft; attributable to a specific flight; or shared. Fuel, pilot salaries, airport costs and cabin crew are flight-variable. Insurance, aircraft lease and aircraft depreciation are aircraft-fixed. It explicitly distinguishes costs that scale with seats (cabin crew, passenger service) from those that do not, and separates passenger cost from cargo cost for mixed operations using a causality-of-cost-occurrence test. |
| Constraint | Confirms the FLIGHT_ATTRIBUTABLE / FLEET_FIXED / PERIOD three-bucket structure. Adds a requirement AMS had not explicitly stated: for a mixed passenger-cargo aircraft, <b>passenger and cargo cost must be separated by causality before unit cost is computed</b> — otherwise belly cargo margin is systematically wrong. <i>See steps 0.2, 2.6, 5.9.</i>                                                                                                                                                    |

## R10 — CORSIA thresholds, exemptions and compliance periods are fixed numbers

**MRV is mandatory for all international operators from 1 January 2019**, independent of participation in offsetting. Operators above **10,000 tonnes CO<sub>2</sub> per year** on international flights must offset. Aircraft below **5,700 kg MTOM** are exempt.

**New entrants are exempt for three years**, or until their emissions reach 0.1% of total 2020 international CO<sub>2</sub>, whichever comes first.

**Compliance periods are three years each:** 2021–2023, 2024–2026, 2027–2029, 2030–2032, 2033–2035.

Offsetting requires **cancelling CORSIA Eligible Emissions Units** equal to the total final requirement for the period.

Five approved fuel-use monitoring methods exist; operators may qualify to use the ICAO **CERT** estimation tool.

Baseline is the 2019–2020 average; from 2024–2035 the obligation is 85% of 2019 CO<sub>2</sub> emissions.

|            |                                                                                                                                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | ICAO <b>Assembly Resolution A41-22</b> and CORSIA FAQ (December 2022); IATA CORSIA programme page; ICAO <b>Annex 16 Volume IV</b> ; Environmental Technical Manual Doc 9501 Volume IV.                                                                                                                  |
| Requires   |                                                                                                                                                                                                                                                                                                         |
| Constraint | All of these are <b>versioned reference data</b> , never code constants. The compliance period mapping must be a table keyed by year. AMS records per-flight fuel because CORSIA MRV requires exactly that, and stores the applicable monitoring method per operator. <i>See steps 0.7, 4.13, 4.14.</i> |

## R11 — EU ETS is priced per tonne and its scope is extending

Aviation has been in the EU ETS since **2012**.

Allowance prices exceeded **EUR 80 per tonne** in 2022 and 2023, producing an estimated total aircraft-operator cost of approximately **EUR 1.8bn in 2022 and EUR 2.1bn in 2023**.

For **2024–2026** the ETS cost was estimated at **4–6% of airlines' total annual operating costs**.

Scope extends to flights **departing the EEA to destinations no further than 5,000 km** from the largest aerodrome in the Union.

Exemption thresholds: commercial operators below **243 flights per period for three consecutive four-month periods**, or below **10,000 t CO<sub>2</sub>/year**.

**CORSIA deduction:** where both apply, CORSIA is deducted to avoid double carbon pricing.

**Non-CO<sub>2</sub> MRV applies from 1 January 2025**, computing CO<sub>2e</sub> per flight from flight, aircraft and fuel properties, performance and weather data.

**SAF** may be treated with a zero emissions factor, with the resulting benefit estimated at 0.5 Mt in 2024 rising to 1.7 Mt in 2026.

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | European Commission, <i>General guidance for Aircraft Operators — EU ETS</i> ; EASA EU Emissions Trading System page; European Commission Q&A on the ETS review, 17 July 2026.                                                                                                                                                                                                                                                               |
| Requires   |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Constraint | At EUR 85/tonne, each tonne allocated to ETS costs a real amount, so the deduction logic must be exact — it is implemented and property-tested in <code>carbon.js</code> . SAF has a computable monetary value (emissions avoided × allowance price), which makes SAF procurement a financially optimisable decision rather than a compliance checkbox. That value must be visible in the sourcing module. <i>See steps 0.7, 4.13, 4.15.</i> |

R12 — PostgreSQL row locking gives exactly the semantics the control needs

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | PostgreSQL documentation, §13.2 <i>Transaction Isolation</i> (all current versions; semantics unchanged since 8.1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Requires   | Read Committed is the PostgreSQL default. On <code>SELECT ... FOR UPDATE</code> , when a target row has already been updated by a concurrent transaction, <i>"the would-be updater will wait for the first updating transaction to commit or roll back"; "If the first updater rolls back, then its effects are negated and the second updater can proceed... If the first updater commits, the second updater will... attempt to apply its operation to the updated version of the row."</i> And decisively: <i>"For the case of SELECT FOR UPDATE and SELECT FOR SHARE, this means that it is the <b>updated version of the row</b> that is locked and returned to the client."</i> |
| Constraint | This is the entire proof. The second transaction is not merely blocked — it receives the row <b>as the first transaction left it</b> , with the committed commitment already deducted. The availability check the second transaction then performs is against post-commit reality, not a stale snapshot. Therefore the race in the specification's Part II P2 is unrepresentable at the database level.                                                                                                                                                                                                                                                                               |

AN IMPORTANT CONSEQUENCE: DO NOT USE SERIALIZABLE HERE

The same documentation states that PostgreSQL achieves true serializability through **predicate locking**, which identifies and flags dependencies and causes serialization errors requiring the application to retry the whole transaction. For a control that must return a clean `BUDGET_EXCEEDED` to an officer, an infrastructure-level serialization failure is the wrong answer — it leaks a database concern into a business workflow and requires every caller to implement retry. Read Committed plus explicit `FOR UPDATE` gives the correct behaviour directly.

R13 — Into-plane fuel cost is negotiated separately from the commodity

|            |                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | IATA, <i>Standard Into-Plane Fueling Service Levels and Safety*</i> (v1.00)                                                                                                                                                                                                                                                                                                                           |
| Requires   | Into-plane providers operate to defined service levels, with fuel quality governed by ICAO Annexes 6 and 14 provisions on fuelling with passengers on board, embarking or disembarking. Service levels distinguish stations where full into-plane capability covering all routine and alternate requirements is available from those where it is not.                                                 |
| Constraint | Fuel price per kg must be captured <b>per delivery note</b> , per station, per date — never as a period average. An airline's choice of airport is driven by into-plane cost as much as by distance or demand, and a monthly average destroys exactly the signal a network planner needs. This is also why fuel uplift is the primary record rather than the invoice. <i>See steps 3.6, 3.7, 4.5.</i> |

R14 — Incumbent systems leave the cost-per-flight gap open

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source     | Vendor product documentation surveyed: Lufthansa Systems <b>NetLine/Sched</b> , <b>NetLine/Ops</b> , <b>NetLine/Crew</b> ; PDC <b>Airline Suite</b> (FlightTime, FlightOps, FlightCrew, FlightCharter, GroundCost, FlightBag); AIMS Airline Software <i>Commercial Planning</i> and <i>Crew Management</i> ; SOMA Software Flight Operations; SmartOPS aviation modules for Odoo.                                                                                                                               |
| Requires   | NetLine/Ops includes a "Tactical Profitability Evaluation Model" that lets planners use a cost and revenue calculation engine in the daily scheduling workspace rather than disconnected spreadsheets. FlightCost & Revenue Analysis in AIMS offers user-definable cost elements — airport charges, crew costs, aircraft depreciation — broken down per flight, per period, region or fleet. PDC GroundCost manages handling contract and invoice control with budget forecasting.                              |
| Constraint | The incumbent answer is a <b>profitability module bolted onto a scheduling tool</b> , computing estimates in the planning workspace. That is genuinely useful and it is not what AMS does. AMS computes <i>actual</i> cost from primary records after the fact, and the planning estimate is a separate digital-twin function (specification Part VII F1). Conflating a planning estimate with an actual cost is how an airline ends up with two numbers and no reconciliation. <i>See steps 2.6, 5.9, 6.7.</i> |

## PART III

## Research Findings That Change the Design

Research is only useful if it changes something. Three findings from Part II required design corrections to the specification as written. They are recorded here rather than silently applied.

### 3.1 Finding 1 — A MEL deferral cannot cure an AD, and must not be offered as if it could

#### SOURCE

EASA CS-GEN-MMEL: *"The MEL cannot deviate from Airworthiness Directives or any other additional mandatory requirements."*

**What the specification says.** AMS keeps ADs and SBs on separate compliance tracks, because applying a Service Bulletin does not make an operator compliant with an Airworthiness Directive. That is correct and necessary.

**What research adds.** The relationship is stronger than "separate". It is **subordinate**. The MEL is legally incapable of addressing an AD, so a system that allows an operator to defer an AD through the MEL workflow is offering an unlawful remedy. Worse, it invites the exact wrong operational response — grounding an aircraft unnecessarily and teaching users to treat the two as interchangeable.

**Design change.** The compliance engine must enforce a hard direction: **MEL deferral cannot clear an AD, and an outstanding AD cannot be deferred through the MEL path.** The dispatch gate must surface outstanding ADs as non-deferrable, and the interface must say why. This is implemented in step 3.9 and tested in step 12.4.

### 3.2 Finding 2 — ADs are organised by ATA chapter, and AMS had no such field

#### SOURCE

14 CFR Part 39 ADs carry an "ATA chapter" field. Observed: **53** Fuselage, **34** Navigation, **32** Landing Gear, **22** Autoflight, **49** APU, **28** Fuel, **21** Air Conditioning, **24** Electrical Power, **26** Flight Controls, **30** Ice and Rain Protection, **31** Instruments, **36** Pneumatic, **55** Powerplant, **25** Cabin Equipment and Furnishings, **23** Communications, **35** Oxygen, **38** Icing, **27** Landing Gear Control, **29** Hydraulics, **33** Navigation Lights.

**Why it matters operationally.** ATA chapter is how technical and compliance staff actually search. An airworthiness engineer asking "what is outstanding on landing gear?" does not think in AD reference numbers — they think in ATA 32. Without the field, every such query is a full-text scan of the AD registry.

**Design change.** Add `ATA_CHAPTERS` to `shared/src/domain.js` and `ata_chapter` to the `adsb_registry` table, indexed. Add ATA chapter as a grouping axis on the Technical portal compliance board and on the dispatch gate output.

```
// Add to shared/src/domain.js (step 1.3)
export const ATA_CHAPTERS = [
  '21', '22', '23', '24', '25', '26', '27', '28',
  '29', '30', '31', '32', '33', '34', '35', '36',
  '38', '45', '46', '47', '48', '49', '51', '52',
  '53', '54', '55', '56', '77',
] as const;
```

### 3.3 Finding 3 — The MEL "day of discovery excluded" rule is an off-by-one

SOURCE

EASA CS-GEN-MMEL: Category B — "within three (3) calendar days, excluding the day of discovery". Category C — ten days, same exclusion. Category D — 120 days, same exclusion.

What the shipped code does. `melStatus()` computes `daysRemaining = floor((rectifyByMs - nowMs) / 86_400_000)` . For a Category B defect discovered at 09:00 on Monday, the regulatory deadline is end of Thursday — three calendar days after Monday, not Monday plus three 24-hour periods. A flight operating Friday morning would be dispatched against a deferral the system considers valid but the regulation considers expired.

Design change. Recompute CAT B/C/D deadlines as a calendar-date operation anchored on the discovery *date*, plus the category day count, truncated to the end of that day. Category A, which specifies no standard interval, must not be inferred from a day count at all.

```
// Corrected deadline logic (step 3.10)
export const MEL_CATEGORIES = {
  A: { days: null,      note: 'no standard interval specified' },
  B: { days: 3,    note: '3 calendar days excluding day of discovery' },
  C: { days: 10,   note: '10 calendar days excluding day of discovery' },
  D: { days: 120,  note: '120 calendar days excluding day of discovery' },
};

// CAT B on 2026-09-28 (Monday) => due end of 2026-10-01 (Thursday),
// NOT 2026-10-01T09:00 - the discovery day is excluded entirely.
function rectifyDeadline(category, discoveredAt) {
  const spec = MEL_CATEGORIES[category];
  if (!spec || spec.days === null) return null; // CAT A
  const discoveryDate = new Date(discoveredAt);
  const due = new Date(discoveryDate);
  due.setUTCDate(due.getUTCDate() + spec.days); // day of discovery excluded
  due.setUTCHours(23, 59, 59, 999);           // valid to END of that day
  return due.getTime();
}
```

3.4 Corrections not made, and why

| Finding                                                                                           | Why not changed                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fuel is commonly described as 25–35% of airline cost (R7 shows labour higher in the US aggregate) | Both are true for different carriers. AMS computes both from primary records per tenant, so no assumption is baked in. The <i>design</i> implication — crew cost is flight-attributable, not a period cost — is already implemented. |
| Non-CO <sub>2</sub> effects MRV applies from 1 January 2025 (R11)                                 | Real requirement, but it needs weather and performance data AMS does not hold in Phase 1. Deferred to the digital-twin work (specification Part VII F1) where the inputs exist. Recorded as open decision O-9.                       |
| Cockpit voice and flight data recorder retrieval requirements in newer Annex 6 amendments (R4)    | Safety and equipment compliance, outside finance, fleet and procurement scope. Recorded for the Phase 2 Safety and Risk module.                                                                                                      |
| Interline and BSP settlement complexity (IATA B16)                                                | Revenue accounting is genuinely complex and is not in the Wave 1 scope. <code>revenue_cents</code> is a single per-flight figure with a documented provenance, so the model is ready when settlement is built.                       |

PART IV

# Wave 0 — Foundation SHIPPED

Wave 0 is the highest-risk wave because every subsequent wave inherits its assumptions. It is complete. Re-running these steps is a regression check.

0.1

Scaffold the monorepo

✓ SHIPPED

ACTION

Create the AMS repository structure with shared contracts, backend, tests and docs as separate concerns under one package.json.

```
# From the workspace root
mkdir ams; cd ams
mkdir shared/src backend/src/config backend/src/db backend/src/audit backend/src/middleware
mkdir backend/src/modules/finance backend/src/modules/fleet backend/src/modules/fuel
mkdir backend/src/modules/procurement backend/migrations backend/tests frontend/src docs

npm init -y
npm pkg set type=module name=ams private=true
npm i -D typescript vitest @types/node
```

VERIFICATION

```
node -e "console.log(process.env.NODE_ENV ?? 'dev')" # must print 'dev' - confirms ESM resolution
ls shared/src backend/src
```

OUTPUT

Directory tree with ESM "type": "module" configured.

EFFORT

XS

0.2

Domain constants — cost taxonomy and departments

✓ SHIPPED

ACTION

Write `shared/src/domain.js` : every state, department, cost category, allocation driver, role and carbon constant in the platform.

```
# The three-bucket cost taxonomy from R7 / R9
export const FLIGHT_ATTRIBUTABLE_CATEGORIES = [
  'fuel', 'crew_flight', 'crew_cabin', 'crew_training',
  'maintenance_direct', 'maintenance_reserve', 'airport_charges',
  'navigation_charges', 'catering', 'handling', 'de_icing',
];
export const FLEET_FIXED_CATEGORIES = [
  'lease_aircraft', 'insurance_aircraft', 'lease_spares',
  'crew_admin', 'distribution',
];
// Period costs are EXCLUDED from CASK. See R7 — this boundary is
// the single largest determinant of whether AMS CASK is comparable.
export const PERIOD_CATEGORIES = [
  'sg_a', 'g_a', 'finance_cost', 'tax',
  'depreciation_other', 'head_office',
];
```

VERIFICATION

| Check                                                                                | Expected                                                          |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <code>COST_CATEGORIES.length === FLIGHT.length + FLEET.length + PERIOD.length</code> | <code>true</code> — no category unclassified                      |
| <code>DEPARTMENTS.length === 16</code>                                               | <code>true</code> — R6 mandates the accountable-manager structure |
| Every department has an entry in <code>DEPARTMENT_PHASE</code>                       | 16 of 16                                                          |

RESEARCH

R6 (accountable managers → 16 departments) · R7 and R9 (three-bucket taxonomy). Correction from Part III.2 applies at step 1.3, not here, because ATA chapters belong to the fleet domain.

EFFORT

S

0.3

Money module — integer cents and exact allocation

✓ SHIPPED

ACTION

Write `shared/src/money.js` . All monetary arithmetic in AMS happens here and nowhere else.

- `cents()` — string-based parse, never via a float.
- `assertSafeInteger()` — rejects floats at every entry point.
- `allocate()` — largest-remainder, parts sum exactly to the total.
- `assertReconciles()` — throws rather than leaking a difference.

VERIFICATION

```
npx vitest run tests/money.test.js
# 17 tests. The two that matter most:
#   - one cent × 1,000,000 = exactly 1,000,000 cents
#   - split US$87,431.66 across 400 flights, parts sum EXACTLY
```

EFFORT

S

0.4

Metrics module — traffic and unit economics

✓ SHIPPED

ACTION

Write `shared/src/metrics.js`: ASK, RPK, load factor, CASK, RASK, break-even load factor, roll-up, utilisation.

CRITICAL DETAIL FROM R8

Roll-up must recompute ratios from summed numerators and denominators, never average per-flight ratios. Averaging load factors weights a 300 km hop equally with a 6,000 km leg, which is a classic modelling error and there is a test asserting the difference.

```
npx vitest run tests/metrics.test.js    # 12 tests
```

RESEARCH

R8 — ASK is available capacity, RPK is revenue demand.  $PLF = RPK \div ASK$ .

EFFORT

S

0.5

Costing engine — the spine

✓ SHIPPED

ACTION

Write `shared/src/costing.js`: flight cost build-up, fleet-fixed allocation, marginal vs full cost, route aggregation and route verdict.

- `buildFlightCost()` — categorises every line; throws on an unrecognised category rather than dropping it silently.
- `allocateFleetFixed()` — six explicit drivers, self-checking with `assertReconciles()`.
- `flightPnl()` — direct, allocated, marginal and full cost.
- `estimateMarginalSeatCost()` — the number most systems never compute and which decides route viability.
- `routeVerdict()` — returns the verdict plus both levers in comparable ppm.

VERIFICATION

```
npx vitest run tests/costing.test.js    # 23 tests
```

RESEARCH

R7 (fuel 16.41% vs labour 34.31% — so crew is flight-attributable) · R9 (three buckets, causality of cost occurrence).

EFFORT

M

0.6

Airworthiness — LLP, AD/SB, MEL

✓ SHIPPED

ACTION

Write `shared/src/airworthiness.js`: LLP controlling-limit arithmetic, AD/SB applicability, dispatch gate, MEL status, check planning.

THE CORE IDEA

Normalise all three life counters to parts-per-million of their *own* limits and take the minimum. A part with 50% cycle life and 4% calendar life has CALENDAR as its controlling limit — and an engineer watching only cycles will fly it into the ground.

VERIFICATION

```
npx vitest run tests/domain-logic.test.js # 45 tests across airworthiness, fuel, carbon
```

RESEARCH

R3 (AD structure) · R2 (MEL categories). Two corrections from Part III land in Wave 3, steps 3.9 and 3.10 — this step shipped the correct shape but not the corrected day-count logic.

EFFORT

M

0.7

Fuel and carbon modules

✓ SHIPPED

ACTION

`shared/src/fuel.js` — burn reconciliation, reserve compliance, tankering economics, variance anomaly detection.

`shared/src/carbon.js` — CO<sub>2</sub> from fuel, SAF zero-factor, CORSIA obligation and compliance periods, EU ETS allocation with the overlap deduction.

VERIFICATION

```
npx vitest run # 118 total
# Specifically assert: emissions are counted EXACTLY ONCE across
# all five regime-applicability combinations (R10 / R11).
```

RESEARCH

R10 (CORSIA thresholds, three-year periods) · R11 (EU ETS scope, price, CORSIA deduction, SAF) · R13 (into-plane cost is per delivery note).

EFFORT

M

0.8

Fuel plan shape — align to R1

✓ SHIPPED

ACTION

Shape `FuelPlan` around the nine prescribed components rather than the intuitive three.

```
// Shipped shape – already correct against R1:
{ taxiFuelKg, tripFuelKg, contingencyKg, alternateKg, finalReserveKg,
  etopsReserveKg, upliftKg, priceCentsPerKg }
// R1 additionally names: EXTRA fuel (delays / operational constraints)
// and DISCRETIONARY fuel (pilot-in-command). Add at step 3.6.
```

VERIFICATION

```
npx vitest run tests/domain-logic.test.js -t "reserves"
```

RESEARCH

R1 — UK CAA SPO.OP131. Final reserve for turbine aeroplanes is 30 minutes at 1,500 ft above the alternate, ISA.

EFFORT

XS (deferred extension)

0.9

Audit hash chain

✓ SHIPPED

ACTION

Write `backend/src/audit/chain.js`: canonical serialisation, chain hashing, full-chain verification, per-tenant verification, anomaly detection.

VERIFICATION

```
npx vitest run tests/audit-costauthority.test.js # 21 tests
# Assert the four attacks are ALL detected:
#   edit · delete · reorder · swap recorded actor
# Assert one tenant's corruption does not invalidate another's.
```

EFFORT

S

0.10

Cost authority — the control core

✓ SHIPPED

ACTION

Write `backend/src/modules/finance/cost-authority.js`: availability arithmetic, row locking, authorisation, pipeline reservation, settlement, release, freeze, batch leg authorisation.

RESEARCH BASIS

R12 — PostgreSQL Read Committed semantics: "it is the updated version of the row that is locked and returned". This is why the control works and why SERIALIZABLE must not be used.

VERIFICATION

```
npx vitest run tests/audit-costauthority.test.js -t "availability"
# Concurrency proof requires PostgreSQL – see step 12.2
```

EFFORT

M

**0.11 Migrations 001–002 with RLS and generated columns**

✓ SHIPPED

**ACTION**

Tenancy, identity, audit with immutability triggers; then budgets, aircraft, commitments with a generated availability column.

```
-- The generated column is the structural reason AMS cannot overspend.
ALTER TABLE budget_lines
  ADD COLUMN available_cents BIGINT
  GENERATED ALWAYS AS (
    budgeted_cents + revised_cents + released_cents
    - committed_cents - pending_cents - expended_cents - frozen_cents
  ) STORED;

ALTER TABLE budget_lines
  ADD CONSTRAINT budget_lines_non_negative
  CHECK (available_cents >= 0);

-- Audit immutability enforced by the database, not by convention.
CREATE TRIGGER audit_logs_no_update BEFORE UPDATE ON audit_logs
  FOR EACH ROW EXECUTE FUNCTION ams_forbid_audit_mutation();
```

**VERIFICATION**

```
docker compose up -d db
node backend/knex-cli.js migrate
psql -h localhost -U ams -d ams -c "\d budget_lines"      # confirm available_cents is GENERATED
psql -h localhost -U ams -d ams -c "UPDATE audit_logs SET action='x' WHERE id=1"
# MUST FAIL with restrict_violation - this is the proof
```

**EFFORT****M****0.12 Type-check gate and CI baseline**

✓ SHIPPED

**ACTION**

`tsconfig.json` with `allowJs`, `checkJs`, `noEmit`, used as a CI gate only. No build output is ever produced.

**VERIFICATION**

```
npx tsc --noEmit # 0 errors
```

**OUTPUT WORTH RECORDING**

On first run the gate reported 47 defects, including two genuine source bugs the unit tests missed: a `ReferenceError` in check planning, and an unreachable verdict branch. This is the empirical justification for ADR-001 and it is recorded in specification Part XVIII.

**EFFORT****XS**

PART V

Wave 1 — Platform, Identity and Portals WEEK 2

Wave 1 turns the pure domain logic into an authenticated, tenant-isolated, audited API with five working portals. It adds no new financial logic — that shipped in Wave 0.

1.1 Config module with Zod-validated environment

ACTION

Parse and validate the environment at boot; refuse to start on an invalid configuration.

```
mkdir -p backend/src/config
# backend/src/config/index.js
import { z } from 'zod';
const env = z.object({
  NODE_ENV: z.enum(['development', 'test', 'production']).default('development'),
  PORT: z.coerce.number().default(4000),
  DB_HOST: z.string().default('127.0.0.1'),
  DB_NAME: z.string().default('ams'),
  DB_USER: z.string().default('ams'),
  DB_PASSWORD: z.string(),
  JWT_SECRET: z.string().min(32),           // refuse short secrets
  JWT_REFRESH_SECRET: z.string().min(32),
  REDIS_URL: z.string().optional(),
}).safeParse(process.env);
if (!env.success) { throw new Error('Invalid environment configuration'); }
export const config = env.data;
```

VERIFICATION

```
JWT_SECRET=short node backend/src/server.js
# MUST refuse to boot - not silently default
```

EFFORT

XS

## 1.2 Migration 003 — departments, roles, portal registry

### ACTION

Seed the 16 departments with phase gating, plus the role catalogue and accountable-manager posts.

```
# backend/migrations/003_departments_roles.js
export const up = async (knex) => {
  await knex.raw(`CREATE TABLE departments (
    id SERIAL PRIMARY KEY,
    tenant_id TEXT NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,
    code TEXT NOT NULL, name TEXT NOT NULL,
    phase SMALLINT NOT NULL DEFAULT 1 CHECK (phase IN (1,2,3)),
    accountable_role TEXT, -- R6: legal post this department must fill
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    UNIQUE (tenant_id, code)
  )`);
  // ... plus RLS policy, as in migration 001
};
```

### VERIFICATION

```
psql -c "SELECT code, phase, accountable_role FROM departments ORDER BY phase, code"
# 16 rows; 5 at phase 1; accountable_role populated for the 6 mandated posts
```

### RESEARCH

**R6** — the six appointed posts (CEO, flight operations, aircraft, safety, quality, security) are a regulatory requirement, not an org choice.

### EFFORT

S

## 1.3 Add ATA chapter codes to the domain RESEARCH

### ACTION

Add `ATA_CHAPTERS` to `domain.js`. This is correction 2 from Part III.2 — discovered in research, required because ADs are organised by ATA chapter and engineers search by ATA chapter.

```
# Append to shared/src/domain.js (full list in Part III.2)
export const ATA_CHAPTERS = [
  '21','22','23','24','25','26','27','28',
  '29','30','31','32','33','34','35','36',
  '38','45','46','47','48','49','51','52',
  '53','54','55','56','77',
];
export const ATA_CHAPTER_NAMES = {
  '21':'Air Conditioning', '22':'Autoflight', '24':'Electrical Power',
  '26':'Flight Controls', '28':'Fuel', '32':'Landing Gear',
  '34':'Navigation', '49':'APU', '53':'Fuselage',
  '55':'Powerplant',
};
```

### VERIFICATION

```
node -e "import('./shared/src/domain.js').then(d=>console.log(d.ATA_CHAPTERS.length))" # 27
```

### RESEARCH

**R3** — ATA chapter codes observed directly in 14 CFR Part 39 ADs: 53 Fuselage, 34 Navigation, 32 Landing Gear, 22 Autoflight, 49 APU, 28 Fuel, 21 Air Conditioning, 24 Electrical Power, 26 Flight Controls, 30 Ice and Rain, 31 Instruments, 36 Pneumatic, 55 Powerplant.

### EFFORT

XS

### 1.4 Database and session bootstrap

ACTION

Knex instance, per-request transaction wrapper, and the RLS tenant binding.

```
# The three layers of tenant isolation begin here
export function withTenant(db, tenantId, fn) {
  return db.transaction(async (trx) => {
    // Layer 3 - the database. RLS engages from this statement onward,
    // so even a raw query with no tenant filter returns nothing.
    await trx.raw('SELECT set_config($1,$2,true)', ['app.tenant_id', tenantId]);
    // Layers 1 and 2 are middleware and repository scoping, layered on top.
    return fn(trx);
  });
}
```

VERIFICATION

```
# In psql, as the application role:
BEGIN; SELECT set_config('app.tenant_id','tenant-a',true);
SELECT count(*) FROM budget_lines WHERE tenant_id='tenant-b'; # must be 0 even for the owner
```

EFFORT

S

### 1.5 Authentication — JWT with refresh rotation

ACTION

Login, short-lived access token, rotating refresh token with replay detection, session revocation.

- Access token 15 minutes; refresh 7 days, rotated on every use.
- A reused refresh token revokes the entire session — that is what replay detection means.
- Bearer token carries `sub`, `tenant_id`, `roles`.

```
curl -X POST localhost:4000/api/v1/auth/login \
  -H 'content-type: application/json' \
  -d '{"email":"cfo@acmeair.co.zw","password":"..."}'
# → { accessToken, refreshToken, tenant, roles, departments }
```

VERIFICATION

| Case                          | Expected                                       |
|-------------------------------|------------------------------------------------|
| Refresh used twice            | Second use returns 401 and revokes the session |
| Access token after 16 minutes | 401 — expiry is enforced, not advisory         |
| Revoked session               | 401 immediately                                |
| Wrong tenant in request body  | Ignored — tenant comes from the token only     |

EFFORT

M

1.6

Accountable-manager appointments

RESEARCH

ACTION

Model the appointment of the six legally mandated posts, including regulatory approval state.

```
# Migration 004 - accountable_managers
# Posts required per R6: ceo · flight_ops · aircraft · safety_officer
#                               quality_manager · security_manager

await knex.raw(`CREATE TABLE accountable_managers (
  tenant_id TEXT NOT NULL,
  post TEXT NOT NULL CHECK (post IN (
    'ceo','flight_ops','aircraft','safety_officer',
    'quality_manager','security_manager')),
  user_id TEXT NOT NULL REFERENCES users(id),
  authority TEXT NOT NULL,           -- e.g. CAAZ, SA-CAA, EASA
  approved_at TIMESTAMPTZ, approved_by TEXT,  -- R6: must be APPROVED, not merely filled
  valid_from DATE NOT NULL, valid_until DATE,
  PRIMARY KEY (tenant_id, post)
)`);
```

VERIFICATION

```
curl localhost:4000/api/v1/compliance/accountable-managers
# MUST flag any post with no APPROVED holder - an operator cannot hold
# or renew its AOC without evidencing these six (R6).
```

RESEARCH

R6 — SA Civil Aviation Regulations 127.06.2(5); corroborated by Mauritius DoC AOC Guidance aligned to ICAO Doc 8335 and EU 2018/1139.

EFFORT

S

## 1.7 Crew licence, medical and training currency RESEARCH

### ACTION

Three separate expiry fields on the crew record — not one combined validity date.

```
// ALREADY IN MIGRATION 001 - verify the separation holds:
license_number      TEXT,
license_type        TEXT,  -- ATPL · MPL · ATPL(A) · CABIN
license_valid_until  DATE,
medical_valid_until  DATE,  -- SEPARATE expiry (R5)
recurrent_training_valid_until DATE, -- SEPARATE expiry (R5)
```

### WHY THREE FIELDS, NOT ONE

A single `valid_until` would let an aircraft be crewed with a lapsed medical but valid licence, because the combined date would still be in the future. The three expire on different cycles and must be checked independently.

### VERIFICATION

```
curl localhost:4000/api/v1/auth/licence
# Returns each expiry separately with days remaining.
```

### RESEARCH

R5 — ICAO Annex 1 Personnel Licensing; CAAZ licenses personnel and approves training organisations per ICAO and national requirements, alongside AFCAC.

### EFFORT

S

## 1.8 Segregation-of-duties evaluator

### ACTION

Centralise SoD rules as a pure function invoked from every state transition — never in the controller.

```
// backend/src/middleware/sod.js - invoked INSIDE the transition
export const SOD_RULES = {
  SELF_APPROVAL: (ctx) => ctx.actorId === ctx.authoredBy,
  /* R6 - evaluator cannot evaluate a procurement they authored */
  EVALUATOR_IS_AUTHOR: (ctx) => ctx.evaluatorId === ctx.authorId,
  AWARD_APPROVER_IS_RECOMMENDER: (ctx) => ctx.approverId === ctx.recommendedBy,
  PAYMENT_RELEASER_IS_ORIGINATOR: (ctx) => ctx.releaserId === ctx.invoiceCreatedBy,
  SUPPLIER_BIDS_OWN_EVENT: (ctx) => ctx.bidderTenantId === ctx.eventOwnerTenantId,
  PERIOD_CERTIFIER_NO_AUTHORITY: (ctx) => !ctx.roles.includes('controller'),
};
```

### VERIFICATION

```
curl -X POST .../cost-authority/authorise -H 'Authorization: Bearer <same-user-token>'
# 422 SOD_SELF_APPROVAL - and no ledger row written
```

### EFFORT

S

## 1.9 Audit middleware writing to the chain

### ACTION

Every mutation writes an audit entry inside the same transaction as the business change, hashed to the previous entry.

### VERIFICATION

```
curl localhost:4000/api/v1/audit-logs/verify
# { valid: true, entriesChecked: N, brokenAt: null }
psql -c "UPDATE audit_logs SET action='tampered' WHERE id=1"
curl localhost:4000/api/v1/audit-logs/verify
# { valid: false, brokenAt: 1 } - but the UPDATE above should have FAILED
# at the trigger. If it succeeded, the trigger is missing (see step 0.11).
```

### EFFORT

S

## 1.10 Idempotency middleware

### ACTION

Accept **Idempotency-Key** on all writes; a replay returns the original result rather than repeating the effect.

### VERIFICATION

```
curl -X POST ../cost-authority/authorise -H 'Idempotency-Key: 11111111-1111-1111-1111-111111111111' -d '{...}'
curl -X POST ../cost-authority/authorise -H 'Idempotency-Key: 11111111-1111-1111-1111-111111111111' -d '{...}'
# Second call MUST return the SAME commitmentId with replayed: true,
# and the budget line must be decremented only once.
```

### EFFORT

S

## 1.11 Rate limiting with separate tiers

### ACTION

Three tiers: general API, authentication (tight), and bid submission (generous per user, but isolated so a tender deadline cannot exhaust the API).

```
// Bid submission gets its own limiter and its own queue - R14 hazard:
// a deadline surge must not degrade the whole API.
app.use('/api/v1/auth', authLimiter); // 20 / 10 min
app.use('/api/v1/bids', bidLimiter); // 100 / hour / user
app.use('/api/v1', apiLimiter); // 300 / 15 min
```

### EFFORT

XS

## 1.12 Transactional outbox and dispatcher worker

### ACTION

Side effects written in-transaction, delivered asynchronously with retry and a dead-letter queue.

```
docker compose up -d db redis
node backend/src/workers/index.js
psql -c "SELECT message_type, count(*) FROM outbox WHERE published_at IS NULL GROUP BY 1"
# MUST drain to zero. A growing queue is a SEV-2 alert.
```

### VERIFICATION

```
psql -c "DELETE FROM outbox WHERE message_type='test' AND attempts > 3"
# Row lands in the dead-letter table, not silently dropped
```

### EFFORT

M

## 1.13 Zod request contracts

### ACTION

Every route declares a Zod schema, applied before any business logic. In a JavaScript codebase this is the type system.

```
// shared/src/schemas.js
export const AuthoriseCost = z.object({
  tenantId: z.string().min(1),
  lineId: z.string().uuid(),
  amountCents: z.number().int().positive().max(1e12), // integer cents, NOT float
  actorId: z.string().uuid(),
  reference: z.string().max(64).optional(),
  idempotencyKey: z.string().uuid().optional(),
}).strict(); // reject unknown keys - a typo must not be silently ignored
```

### VERIFICATION

```
curl -X POST ../authorise -d '{"amountCents": 100.5}'
# 400 - fractional cents rejected at the boundary
```

### EFFORT

S

1.14

Server wiring and health endpoint

ACTION

Assemble the middleware stack in the fixed order documented in specification §13.4. Order is load-bearing: correlation id, then auth, then tenant, then roles, then Zod.

```
curl localhost:4000/api/v1/health
# { status:'ok', db:'up', redis:'up', outboxLag: 0, chainHead: '9f2c...' }
```

VERIFICATION

```
# Middleware order test - confirm tenant binds BEFORE any query:
npx vitest run tests/middleware-order.test.js
```

EFFORT

S

1.15

Wave 1 portals — five functional

ACTION

Build the five Wave 1 portals and eleven honest Phase 2 shells.

| Portal                  | Minimum viable screens for wave exit                                                                                                                                            |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finance & Treasury      | Flight cost explorer · route P&L board · cost authority queue · budget builder                                                                                                  |
| Procurement & Supply    | Vendor registry · sourcing event list · contract register                                                                                                                       |
| Technical / Engineering | Fleet register · component register · AD/SB board · dispatch gate · check planning                                                                                              |
| Executive               | Consolidated P&L · fleet performance · route verdict board · carbon exposure                                                                                                    |
| IT / MIS                | Users · roles · delegations · cost taxonomy · audit chain verification                                                                                                          |
| 11 × Phase 2            | Shell stating what is coming, with role definitions and entitlements. No fake data — a portal showing fabricated figures trains people to trust a screen that will be replaced. |

VERIFICATION

```
npx playwright test # covers the two critical journeys end to end
```

EFFORT

L

1.16 Wave 1 verification gate

ACTION

Run the consolidated Wave 1 gate (Part XIV) and close the wave only if every line passes.

VERIFICATION

```
npm test && npx tsc --noEmit
npx playwright test
npm run security:tenant-isolation    # cross-tenant suite
```

EFFORT

S

PART VI

# Wave 2 — Cost Authority and Finance Operations CRITICAL PATH

WHY THIS WAVE IS THE CRITICAL PATH

Fuel, fleet, contracts, reporting and every downstream module hang off the cost authority and the budget line. Retrofitting commitment control after the fact means rewriting half the platform. This wave is also where the concurrency proof becomes a permanent automated gate rather than a one-off demonstration.

## 2.1 Migration 005 — cost structure and reference data

ACTION

Create cost categories with structural buckets, the cost centre tree, and funds. Seed from the IATA-aligned taxonomy, not invented codes.

```
CREATE TABLE cost_categories (  
  id SERIAL PRIMARY KEY,  
  tenant_id TEXT NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,  
  code TEXT NOT NULL, name TEXT NOT NULL,  
  bucket TEXT NOT NULL CHECK (bucket IN ('FLIGHT_ATTRIBUTABLE', 'FLEET_FIXED', 'PERIOD')),  
  
  -- R7/R9: the bucket is what disqualifies a category from CASK.  
  -- It is a structural property, not a report-time toggle.  
  drives TEXT[] NOT NULL DEFAULT '{}',  
  is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  UNIQUE (tenant_id, code)  
);
```

VERIFICATION

```
psql -c "SELECT bucket, count(*) FROM cost_categories GROUP BY bucket ORDER BY 2 DESC"  
# 11 FLIGHT_ATTRIBUTABLE · 5 FLEET_FIXED · 6 PERIOD  
psql -c "SELECT count(*) FROM cost_categories WHERE bucket IS NULL" # must be 0
```

RESEARCH

R7, R9.

EFFORT

S

## 2.2 Budget versioning and approval

ACTION

Budgets are versioned; only one version per line per year is **ACTIVE**. Revisions never edit in place.

VERIFICATION

```
# Attempt to activate two versions on the same line:  
psql -c "UPDATE budget_versions SET status='ACTIVE' WHERE budget_id=1 AND version=2"  
# MUST be blocked by a partial unique index:  
# CREATE UNIQUE INDEX ON budget_lines (id) WHERE status='ACTIVE'
```

EFFORT

S

## 2.3 Pipeline reservation on submit

### ACTION

Reserve budget capacity at *submission*, not only at approval.

#### WITHOUT THIS, THE QUEUE OVER-COMMITS ITSELF

An approver opening their inbox would see headroom that fifteen pending requests have already reserved, and approve a sixteenth. Subtracting `pending_cents` from availability at submit time is what prevents that. This is a small change with an outsized effect on real-world behaviour.

### VERIFICATION

```
curl -X POST ../requisitions/111/submit -d '{"lines":[{"lineId":"...", "amountCents":100000}]}'
curl localhost:4000/api/v1/budget-lines/<lineId>
# pending_cents must have INCREASED and available_cents DECREASED,
# with no commitment ledger row yet.
```

### EFFORT

S

## 2.4 Cost authority endpoints CRITICAL

### ACTION

Expose the Wave 0 `cost-authority.js` functions as authenticated endpoints. The logic shipped; only wiring remains.

```
POST /api/v1/cost-authority/authorise      # creates a commitment
POST /api/v1/cost-authority/authorise-leg # all-or-nothing across a flight leg
POST /api/v1/cost-authority/settle        # does NOT restore availability
POST /api/v1/cost-authority/release       # variance; restores availability
POST /api/v1/budget-lines/:id/freeze      # reason mandatory, min 10 chars
```

### VERIFICATION

```
curl -X POST ../authorise -H 'Idempotency-Key: <uuid>' \
  -d '{"lineId":"<uuid>", "amountCents":9999999999}'
# 422 BUDGET_EXCEEDED
# { "availableCents": 1000000, "requestedCents": 9999999999,
#   "shortfallCents": ... } ← the UI needs the shortfall to be useful
```

### EFFORT

M

## 2.5 THE CONCURRENCY PROOF CRITICAL

### ACTION

Write the automated test that proves over-commitment is unrepresentable, and wire it as a permanent gate.

```
# tests/integration/concurrency.test.js - requires PostgreSQL
describe('over-commitment under concurrency', () => {
  it('50 concurrent authorisations, headroom for 10 => exactly 10 succeed', async() => {
    // Budget line: available = 1,000,000c ($10,000). Each request asks 100,000c.
    await seedLine({ available: 1_000_000 });
    const results = await Promise.allSettled(
      Array.from({ length: 50 }, () => authorise(100_000));

    const ok      = results.filter(r => r.status === 'fulfilled');
    const denied  = results.filter(r => r.status === 'rejected');

    expect(ok.length).toBe(10); // exactly the headroom
    expect(denied.length).toBe(40);
    denied.forEach(d => expect(d.reason.code).toBe('BUDGET_EXCEEDED'));

    // The ledger must sum to exactly the headroom - no drift, no leakage.
    const line = await readLine();
    expect(line.committed_cents).toBe(1_000_000);
    expect(line.available_cents).toBe(0);
    expect(line.available_cents).toBeGreaterThanOrEqual(0); // never negative
  });
});
```

### THE TEST THAT MAKES THIS REAL

Temporarily remove `.forUpdate()` from `lockBudgetLine()` and re-run. The test **MUST** fail — with 50 commitments succeeding against headroom for 10. A concurrency test that passes without its lock is not testing anything. Wire this into CI as a release gate; it is the single most important test in the project.

### RESEARCH

*R12 — the PostgreSQL semantics that make this work, and the reason **SERIALIZABLE** must not be used instead.*

### EFFORT

S

## 2.6 Flight cost attribution service

### ACTION

Assemble the attributed cost of a flight from primary records, then expose it.

```
GET /api/v1/flights/:id/cost
{
  flightId, registration, flightNumber, date,
  direct:      { fuel: 304000, crew_flight: 185000, ... },    # integer cents
  fleetFixed:  [ { category: 'lease_aircraft', cost: 61200 }, ... ],
  fullCostCents, marginalSeatCostCents, breakEvenLoadFactorPpm,
  caskMicrocents, raskMicrocents, contributionCents
}
```

### VERIFICATION

```
curl localhost:4000/api/v1/flights/<id>/cost
# Assert: sum(direct) + sum(fleetFixed) === fullCostCents
# Assert: marginalSeatCostCents <= fullCostCents  ← the key relationship
# Assert: every value is a safe integer (no floats in the response)
```

### RESEARCH

*R14 — this is the actual-versus-estimate distinction. Vendors compute estimates in the planning workspace; AMS computes actuals from primary records. Conflating them is why two numbers exist with no reconciliation.*

### EFFORT

M

## 2.7 Passenger and cargo separation by causality RESEARCH

### ACTION

For a mixed passenger-cargo aircraft, separate passenger cost from cargo cost *before* computing unit cost.

#### FROM R9

"If an airplane carries passengers and cargo, costs are first divided into passenger or cargo costs based on causality of cost occurrence." Skipping this step makes belly-cargo margin systematically wrong, because cargo consumes payload that passengers would otherwise use and the fuel consequence lands on whichever side is allocated first.

### VERIFICATION

```
curl localhost:4000/api/v1/flights/<id>/cost | jq '.costSeparation'
# { passengerCostCents, cargoCostCents, basis: 'CAUSALITY', unallocated: 0 }
# unallocated MUST be 0 - an unallocated residue means the split is arbitrary
```

### EFFORT

M

2.8

Monthly cash-flow forecast as a closing prerequisite

ACTION

The controller cannot certify a period without a submitted forecast.

```
curl -X POST ../periods/2026-08/certify
# 422 CASH_FORECAST_REQUIRED - the response must name the controller and
# the missing forecast, not merely fail
```

RESEARCH

Discipline carried from the government PFMS specification (SI 144/2019 required monthly cash-flow forecasts) and applied to an airline treasury.

EFFORT

S

2.9

Accounts payable and the multi-way match

ACTION

Supplier invoice matched against purchase order, receipt *and* — for fuel — the delivery note; for crew cost — the duty report.

```
POST /api/v1/invoices/:id/match
# { matched: false, variances: [
#   { type: 'QUANTITY', line: 3, ordered: 3200, received: 3000 },
#   { type: 'PRICE',    line: 1, unit: 95, invoiced: 104 } ] }
```

VERIFICATION

| Case                                        | Expected                                             |
|---------------------------------------------|------------------------------------------------------|
| Invoice with no receipt                     | Payment blocked; exception recorded                  |
| Invoice above ordered quantity              | Blocked above tolerance; partial permitted within it |
| Fuel invoice not matching any delivery note | Blocked — this is a real fuel-theft detection point  |
| Price variance beyond tolerance             | On hold, routed to the purchase order holder         |

RESEARCH

R13 — *per-delivery-note fuel capture is what makes the fourth-way match possible at all.*

EFFORT

L

## 2.10 Payment runs and bank file generation

### ACTION

Build a payment batch, generate an RTGS-format file, and settle.

```
POST /api/v1/payment-runs
curl localhost:4000/api/v1/payment-runs/<id>/file -o payments.rtgs
# Re-submitting the same run MUST NOT create a second settlement (idempotency, R14)
```

### VERIFICATION

```
curl -X POST ../payment-runs/<id>/export -H 'Idempotency-Key: <uuid>'
curl -X POST ../payment-runs/<id>/export -H 'Idempotency-Key: <uuid>'
psql -c "SELECT count(*) FROM payments WHERE payment_run_id='<id>'"
# must be the number of invoices in the run, NOT double
```

### EFFORT

M

## 2.11 Settlement semantics — availability must not be restored

### ACTION

Assert the single most commonly implemented rule incorrectly.

```
npx vitest run tests/costing.test.js -t "payment does not restore"
# THE RULE: settlement decrements committed_cents and increments
# expended_cents. available_cents DOES NOT MOVE. Money has left.
# Only a budget release or a revision increases availability.
```

### WHY THIS MATTERS MORE THAN IT LOOKS

Restoring availability on settlement would let the same money be committed a second time — the exact failure the whole module exists to prevent. A system that gets this wrong passes every functional test and still overspends.

### EFFORT

S

## 2.12 Variance release

### ACTION

Handle a supplier invoicing less than ordered — a legitimate and common event — as an explicit variance, never a silent absorption.

```
curl -X POST ../cost-authority/release \
  -d '{"commitmentId":"...", "releaseCents":45000, "reason":"Supplier short-shipped 320 kg on fuel"}'
# availability RESTORED by 45000; released_cents updated; reason stored
```

### VERIFICATION

```
# Releasing more than is outstanding MUST be refused:
curl -X POST ../release -d '{"commitmentId":"...", "releaseCents":99999999}'
# 422 RELEASE_EXCEEDS_COMMITMENT
```

### EFFORT

S

## 2.13 Treasury freeze with mandatory reason

### ACTION

Ring-fence funds without creating a commitment — security deposits, litigation holds, investigations.

```
curl -X POST ../budget-lines/<id>/freeze \
  -d '{"amountCents":250000, "reason":"Security deposit pending litigation"}'
curl -X POST ../budget-lines/<id>/freeze -d '{"amountCents":250000, "reason":"x"}'
# 422 FREEZE_REASON_REQUIRED - minimum 10 characters. This is a governance
# action that appears in the audit pack; "x" is not an acceptable record.
```

### VERIFICATION

```
curl localhost:4000/api/v1/budget-lines/<id>
# frozen_cents INCREASES, available DECREASES, committed UNCHANGED -
# frozen money is unspendable but was never promised to anyone
```

### EFFORT

S

## 2.14 Devolution and division allocation

### ACTION

Model allocation from a group budget down to divisions, cost centres and — where applicable — regional allocations.

### VERIFICATION

```
psql -c "SELECT SUM(allocated_cents) FROM budget_allocations WHERE parent_line_id='<x>'"
# MUST equal parent's available allocation - no leakage across the tree
```

### EFFORT

M

## 2.15 Finance portal: cost authority queue

### ACTION

The screen an officer actually lives in. Design constraint: availability must be visible *before* submission, not on rejection.

```
GET /api/v1/cost-authority/queue
# Each row: requester · department · category · requested · available ·
#           pipelineReserved · verdict
```

### VERIFICATION

```
npm run test:e2e -- cost-authority
# Assert p95 for the availability endpoint < 200ms - an officer must feel
# it is instant, or they will approve without looking
```

### EFFORT

M

## 2.16 Budget variance reporting

### ACTION

Budget-to-actual with variance, decomposed to the same grain as cost attribution.

### VERIFICATION

```
curl localhost:4000/api/v1/reports/budget-execution?period=2026-08
# variance must reconcile: actual - budget = variance, exactly, in integer cents
```

### EFFORT

S

## 2.17 Backfill and re-run costing

### ACTION

Allow cost to be rebuilt from primary records for a historical period, deterministically.

### VERIFICATION

```
curl -X POST ../flights/cost/build -d '{"period":"2026-01","mode":"DRY_RUN"}'
# Returns the diff WITHOUT writing. Then run for real and diff again -
# the second run must produce zero diff. Non-determinism in costing is a
# bug, not a rounding artefact.
```

### EFFORT

S

2.18 Wave 2 verification gate

ACTION

Close the wave only when the concurrency proof and every Part XIV gate passes.

```
npm test
npx tsc --noEmit
npm run test:integration
npx vitest run tests/integration/concurrency.test.js # THE gate
```

EFFORT

S

PART VII

# Wave 3 — Fleet, Fuel Operations and Stores

## 3.1 Migration 007 — fleet register

ACTION

Aircraft master with registration, MSN, configuration, lease type and utilisation counters.

VERIFICATION

```
psql -c "SELECT registration, serial_number, type_code, seats_installed, mtom_kg
        FROM aircraft WHERE tenant_id='acme'"
# UNIQUE (tenant_id, registration) - a tail number is unique within an airline
```

RESEARCH

R4 — fleet type and MSN are also the keys by which AD applicability is determined at step 3.8.

EFFORT

S

## 3.2 IFRS 16 lease accounting

ACTION

Model the right-of-use asset and matching liability, interest unwinding and depreciation over the term.

VERIFICATION

```
curl localhost:4000/api/v1/aircraft/Z-WAB/lease
# Assert: ROU asset NBV + liability = 0 net of payments, per period
# Assert: interest + depreciation = lease charge, exactly, in integer cents
```

RESEARCH

IFRS 16 requires almost every aircraft lease to be recognised on balance sheet.

EFFORT

M

## 3.3 Component register with serialisation

ACTION

Serial-numbered parts, locations, shelf life, and installation provenance.

VERIFICATION

```
curl localhost:4000/api/v1/components/P%2FN-4471
# { partNumber, serialNumber, installedOn, hoursAtInstall, cyclesAtInstall,
#   llp: { controllingCounter, remaining, status }, adsb: [...] }
```

EFFORT

M

### 3.4 Component event chain — the provenance record

ACTION

Append-only chain of removal, inspection, installation and life accumulation. This is the substrate for external anchoring (specification Part VII F4).

VERIFICATION

```
psql -c "SELECT count(*) FROM component_events WHERE component_id='<sn>'"
# Verify the hash chain over component events exactly as for the audit log
```

EFFORT

M

### 3.5 Fuel delivery note capture RESEARCH

ACTION

Capture uplift at the fuel station, per delivery note, per station, per date, with the into-plane price separated from the commodity.

FROM R13

Price per kg must be captured per delivery note, never as a period average. An airline's choice of airport is driven by into-plane cost as much as distance or demand, and a monthly average destroys exactly the signal a network planner needs.

VERIFICATION

```
curl -X POST ../fuel/uplifts -d '{
  "registration":"Z-WAB", "station":"HRE", "kg":3200,
  "commodityPriceCentsPerKg":78, "intoPlaneFeeCentsPerKg":17,
  "deliveredAt":"2026-03-14T06:12:00Z", "deliveryNote":"DN-88213"
}'
# Assert price == commodity + intoPlane, in integer cents
```

EFFORT

M

### 3.6 Complete the nine-component fuel plan RESEARCH

#### ACTION

Extend `FuelPlan` with the components named in R1 that Wave 0 deferred: extra fuel and discretionary fuel.

```
// Extension to shared/src/fuel.js
export function buildFuelPlan({
  taxi, trip, contingency, destinationAlternate, finalReserve,
  additionalEtop,    // FE ERA critical scenario (R1)
  extra,             // anticipated delays / operational constraints
  discretionary,     // pilot-in-command
}) {
  return {
    components: { taxi, trip, contingency, destinationAlternate, finalReserve,
                  additionalEtop, extra, discretionary },
    // R1: final reserve for turbine-engined aeroplanes is 30 minutes
    // at 1,500 ft above the alternate, ISA conditions.
    finalReserve: { minutes: 30, holding: true, isaConditions: true },
    totalUsable: taxi + trip + contingency + destinationAlternate
                  + finalReserve + additionalEtop + extra + discretionary,
  };
}
```

#### VERIFICATION

```
npx vitest run tests/domain-logic.test.js -t "reserve"
# Assert finalReserve defaults to 30 min for turbine types
# Assert a flight with zero final reserve is FLAGGED, not silently accepted
```

#### RESEARCH

R1 — UK CAA SPO.OP.131 fuel components.

#### EFFORT

S

### 3.7 Burn reconciliation and reserve compliance

#### ACTION

Reconcile uplift to burn, carry the landing reserve forward, and flag a short reserve as a regulatory finding.

#### VERIFICATION

```
curl localhost:4000/api/v1/fuel/reconcile/<flightId>
# { plannedBurnKg, actualBurnKg, varianceKg, variancePpm,
#   reservesKg, expectedLandingFuelKg,
#   reservesMeetRegulatoryMinimum: false,  ← THIS is the finding
#   varianceCostCents }
```

#### EFFORT

M

3.8

Full AD registry structure

RESEARCH

ACTION

Expand the AD model beyond "reference, type, due date" to the full structure observed in real directives.

```
CREATE TABLE adsb_registry (  
  id SERIAL PRIMARY KEY,  
  tenant_id TEXT NOT NULL,  
  reference TEXT NOT NULL,           -- 'EASA AD 2024-0184'  
  kind TEXT NOT NULL CHECK (kind IN ('AD','SB')),  
  authority TEXT NOT NULL,           -- EASA / FAA / CAAZ / ... (R3)  
  ata_chapter TEXT NOT NULL,         -- '32' Landing Gear (Part III.2)  
  aircraft_type TEXT NOT NULL,  
  msn_from BIGINT, msn_to BIGINT,    -- serial range (R3)  
  mods_required TEXT[],              -- 'embodied modification 21202'  
  mods_excluded TEXT[],              -- 'except modification 152569'  
  reason TEXT NOT NULL,              -- the safety finding (R3)  
  sb_reference TEXT,                 -- accomplishment instruction  
  due_cycles BIGINT, due_hours NUMERIC, due_date DATE, -- multiple thresholds (R3)  
  repetitive_interval_cycles BIGINT,  
  repetitive_interval_hours NUMERIC,  
  non_cumulative_tolerance_ppm INT DEFAULT 50000,      -- R3: 5%  
  terminating_action TEXT,           -- CMM repair ending inspections (R3)  
  is_emergency BOOLEAN DEFAULT FALSE,  
  operating_limitation TEXT,         -- emergency ADs (R3)  
  superseded_by TEXT REFERENCES adsb_registry(id),  
  effective_from DATE NOT NULL  
);
```

VERIFICATION

```
curl 'localhost:4000/api/v1/adsb?ataChapter=32&status=DUE_SOON'  
# Must work - engineers search by ATA chapter, not by AD number
```

RESEARCH

R3 — structure observed across 14 CFR Part 39 (79 FR 19848; 87 FR 67354), EASA AD 2026-0125-E, AD 2023-0167R1, AD 2024-0213.

EFFORT

M

### 3.9 MEL can never cure an AD CRITICAL RESEARCH

#### ACTION

Enforce the subordination relationship. This is correction 1 from Part III.1 and it is the most important finding in the research.

#### THE RULE

**EASA CS-GEN-MMEL:** *"The MEL cannot deviate from Airworthiness Directives or any other additional mandatory requirements."*

A MEL deferral is legally incapable of addressing an outstanding AD. A system that allows an operator to defer an AD through the MEL workflow offers an unlawful remedy and teaches the exact wrong operational response.

```
// backend/src/modules/fleet/compliance.js
export function evaluateCompliance(aircraft, registry, melDeferrals) {
  const applicable = registry.map((ad) => adsbApplies(ad, aircraft)).filter(r => r.applies);

  const deferredAdRefs = melDeferrals
    .filter(d => applicable.some(a => a.complianceStatus.reference === d.itemRef))
    .map(d => d.itemRef);

  // REFUSE to treat any AD as satisfied by a MEL deferral.
  // The relationship is subordinate, not parallel.
  for (const ad of applicable) {
    /* eslint-disable-next-line no-param-reassign */
    ad.complianceStatus.deferred = false; // never true, by law (R2)
    ad.complianceStatus.deferralAttempted =
      deferredAdRefs.includes(ad.complianceStatus.reference);
  }

  const overdue = applicable.filter(r => r.complianceStatus.status === 'OVERDUE');
  const illegalAttempts = applicable.filter(r => r.complianceStatus.deferralAttempted);

  return {
    dispatchable: overdue.length === 0,
    overdue,
    // Surfaced separately so the operator sees WHY, not just a block.
    illegalDeferralAttempts: illegalAttempts.map(r => r.complianceStatus.reference),
  };
}
```

#### VERIFICATION

```
npx vitest run tests/fleet-compliance.test.js
# Assert: an AD with an active matching MEL deferral is STILL OVERDUE
# Assert: the deferral attempt is reported as illegal, not honoured
# Assert: a genuine SB may be recorded as applied – and the AD remains open
```

#### RESEARCH

R2 — EASA CS-GEN-MMEL Issue 2, Annex II to ED Decision 2020/012/R.

#### EFFORT

S

### 3.10 MEL rectification with the day-of-discovery exclusion RESEARCH

#### ACTION

Replace the shipped duration arithmetic with calendar-date arithmetic that excludes the discovery day. Correction 3 from Part III.3.

```
export const MEL_CATEGORIES = {
  A: { days: null, note: 'no standard interval specified' },
  B: { days: 3, note: '3 calendar days excluding day of discovery' },
  C: { days: 10, note: '10 calendar days excluding day of discovery' },
  D: { days: 120, note: '120 calendar days excluding day of discovery' },
};
```

#### VERIFICATION — THE SPECIFIC CASE THAT PROVES IT

```
npx vitest run tests/mel.test.js -t "CAT B"
# Category B discovered 2026-09-28 (Monday) 09:00 UTC
# → deadline 2026-10-01 (Thursday) 23:59:59.999 UTC
# NOT 2026-10-01T09:00. The discovery day is excluded entirely.
# A flight at 06:00 Friday MUST be dispatched as EXPIRED.
```

#### RESEARCH

*R2 — Category B 3 days, C 10 days, D 120 days, all "excluding the day of discovery". Category A specifies no standard interval and must not be inferred from a day count.*

#### EFFORT

S

### 3.11 MEL item structure and (M)/(O) symbols RESEARCH

#### ACTION

Model the five MMEL columns and the modification / operating-limitation flags as first-class attributes.

```
// MMEL columns per R2: system & sequence number · rectification interval ·
// number installed · number required for dispatch · remarks/exceptions
{ itemRef: '22-10-1B', ataChapter: '22', category: 'B',
  installedCount: 1, requiredForDispatch: 1,
  requiresModification: true, // the (M) symbol
  requiresOperatingLimitation: true } // the (O) symbol
```

#### WHY (O) IS NOT COMMENTARY

An (O) item imposes an operating limitation that changes what the crew is permitted to do — it is not advisory text. If the deferral carries (O), AMS must surface the limitation on the dispatch brief, not merely record that a deferral exists.

#### RESEARCH

*R2 — CS-GEN-MMEL item table structure and the (M)/(O) symbols.*

#### EFFORT

S

### 3.12 Dispatch gate with ATA chapter grouping

#### ACTION

**Aggregate every compliance item into a single dispatch decision, grouped by ATA chapter.**

#### VERIFICATION

```
curl localhost:4000/api/v1/aircraft/Z-WAB/dispatch-gate
{
  dispatchable: false,
  overdueCount: 1,
  overdue: [ { reference: 'EASA AD 2026-0125-E', ataChapter: '32',
    ataChapterName: 'Landing Gear', status: 'OVERDUE',
    deferralAttempted: true } ], // ← step 3.9
  nextDue: { ataChapter: '34', status: 'DUE_SOON', remainingPpm: 41_666 }
}
```

#### RESEARCH

*R3 (ATA chapter grouping) · R4 (dispatch is a legal act requiring every prerequisite satisfied).*

#### EFFORT

**M**

### 3.13 Check planning across all three counters

#### ACTION

**Compute hours remaining, cycles remaining, calendar remaining, and identify the controlling counter in calendar days at current utilisation.**

#### VERIFICATION

```
curl localhost:4000/api/v1/aircraft/Z-WAB/checks
# Assert controllingCounter is the one with the fewest DAYS remaining,
# not the fewest absolute units – the units are incommensurable
```

#### RESEARCH

*R3 — real AD compliance times: "before exceeding 24,000 flight cycles or 48,000 flight hours, whichever occurs first".*

#### EFFORT

**S**

### 3.14 Stores, cycle counting and Wave 3 gate

#### ACTION

**Spares and catering inventory with append-only movements, cycle counting, then close the wave.**

```
npm test && npx tsc --noEmit && npm run test:e2e
npx vitest run tests/mel.test.js tests/fleet-compliance.test.js
# Both Part III corrections MUST be asserted here
```

#### EFFORT

**M**

PART VIII

# Wave 4 — Procurement and Carbon Compliance

## 4.1 Aviation vendor categories

ACTION

Vendor registry with categories that reflect what actually grounds an aircraft, not generic procurement taxonomy.

| Category        | Special obligation AMS must model                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| FUEL            | Into-plane fee structure · delivery-note reconciliation · price per kg by station and date · hedging linkage                  |
| SPARES          | Serialisation · life-limited status · AD/SB applicability · shelf life · repair vs replace economics · counterfeit risk flags |
| AIRCRAFT_LEASE  | IFRS 16 right-of-use asset and liability · term · residual value · return-condition obligations                               |
| GROUND_HANDLING | Per-sector and per-passenger rates · turnaround commitments · SLA breach credits                                              |
| CATERING        | Cost per passenger · uplift per rotation · wastage · special-meal handling                                                    |
| INSURANCE       | Hull value tied to the asset register · deductibles · coverage period against lease term                                      |

EFFORT

S

## 4.2 Vendor document expiry alerts

ACTION

Track tax clearance, licences and certifications with expiry, and block award against an expired document.

VERIFICATION

```
curl localhost:4000/api/v1/vendors/<id>/eligibility
# { eligible: false, blockers: ['TAX_CLEARANCE_EXPIRED', 'AIR_WORTHINESS_LICENCE_EXPIRED'] }
curl -X POST ../awards -d '{ vendorId: "<expired>" }'
# 422 - an award cannot be recorded against an ineligible vendor
```

RESEARCH

R5 — licensing authority interaction; R6 — approvals must be current, not merely recorded.

EFFORT

S

### 4.3 Procurement plan with plan-linkage gate

ACTION

A sourcing event cannot publish without an approved, value-bearing plan line.

VERIFICATION

```
curl -X POST ../sourcing/<id>/publish # no linked plan line
# 422 PLAN_LINE_REQUIRED - the plan makes procurement visible BEFORE it happens
```

EFFORT

S

### 4.4 Sealed bid submission with receipt hash

ACTION

Bid submission timestamped and content-hashed on receipt; encrypted at rest; visible only to a constituted opening committee after closing.

VERIFICATION

```
curl -X POST ../bids -H 'Idempotency-Key: <uuid>' -F 'document=@bid.pdf'
# { receiptHash: 'a3f9...', submittedAt: '2026-05-04T14:00:03Z' }
# A retried upload MUST return the same receiptHash, not a new bid
```

EFFORT

M

### 4.5 Bid opening with an immutable register

ACTION

Opening requires a constituted committee at quorum. The register is generated, signed and then immutable.

VERIFICATION

| Case                                          | Expected                                  |
|-----------------------------------------------|-------------------------------------------|
| Opening without a constituted committee       | Refused; no register generated            |
| Opening below quorum                          | Refused                                   |
| Amending the register after signing           | Blocked by trigger, as with the audit log |
| Any bidder viewing another bid before opening | 403                                       |

EFFORT

M

## 4.6 Evaluation against declared criteria

### ACTION

Scores must map to a criterion declared in the bidding document. Inventing a criterion at evaluation is prohibited.

### VERIFICATION

```
curl -X POST .../evaluations/<id>/scores \
  -d '{"bidId":"...", "criterionCode":"NOT_IN_TENDER", "score":9}'
# 422 CRITERION_NOT_DECLARED
```

### EFFORT

M

## 4.7 Award with preference application

### ACTION

Record the award with the preference rationale; require completed evaluation, post-qualification and preference.

### VERIFICATION

```
curl -X POST .../awards
# 422 if post-qualification is missing, OR if preference was not applied -
# the rationale must be recorded either way, including "no preference applies"
```

### EFFORT

M

## 4.8 Contract obligation models

### ACTION

Each contract type carries its own obligation model — not a generic amount and date.

### VERIFICATION

```
curl localhost:4000/api/v1/contracts/<id>
# fuel_offtake: { priceFormula, intoPlaneBands[], shortfallPenalty,
#                volumeCommitment, forceMajeure }
# handling:     { perSectorRate, perPaxRate, turnaroundCommitment,
#                slaCredits[] }
# catering:     { perPaxRate, specialMealRate, wastageAllowance }
# aircraft_lease:{ rouAsset, liability, residualValue, returnConditions[] }
```

### RESEARCH

R11 (SAF and fuel cost drivers) · R13 (into-plane structure) · IFRS 16 (lease obligations).

### EFFORT

L

4.9

Per-flight emissions ledger

ACTION

Fuel × ICAO factor → CO<sub>2</sub> per flight, stored on the flight. This is the substrate for everything else in 4.10–4.12.

VERIFICATION

```
curl localhost:4000/api/v1/carbon/flights/<id>
# { fuelKg: 3200, totalCo2Tonnes: 10.112, safKg: 640, safSharePpm: 200000 }
```

RESEARCH

R10 — MRV requires per-flight fuel use for every international flight.

EFFORT

S

4.10

CORSIA obligation determination

RESEARCH

ACTION

Determine obligation from thresholds, exemptions and the new-entrant clock — as versioned data.

| Condition                            | Threshold                                                                 |
|--------------------------------------|---------------------------------------------------------------------------|
| Annual international CO <sub>2</sub> | 10,000 tonnes — above this, obligated                                     |
| Aircraft MTOM                        | 5,700 kg — below this, exempt                                             |
| New entrant                          | Exempt 3 years, or until 0.1% of total 2020 international CO <sub>2</sub> |
| Compliance periods                   | 2021–2023 · 2024–2026 · 2027–2029 · 2030–2032 · 2033–2035                 |
| Baseline                             | 2019–2020 average; 85% of 2019 CO <sub>2</sub> for 2024–2035              |

VERIFICATION

```
curl localhost:4000/api/v1/carbon/obligation?year=2026
# { obligated: false, period: '2024-2026',
#   reason: 'Annual international CO2 9,999t is below the 10000t threshold' }
```

RESEARCH

R10 — ICAO Assembly Resolution A41-22; CORSIA FAQ (Dec 2022); Annex 16 Vol IV.

EFFORT

S

4.11

EU ETS charging with the CORSIA deduction

RESEARCH

ACTION

Allocate each flight's emissions across regimes, deducting CORSIA-covered tonnes from the ETS chargeable quantity. Shipped in `carbon.js`; this step exposes and prices it.

WHY PRECISION MATTERS COMMERCIALLY

Allowance prices exceeded EUR 80/tonne in 2022 and 2023, with total aircraft-operator cost around EUR 1.8bn in 2022 and EUR 2.1bn in 2023, estimated at 4–6% of annual operating cost for 2024–2026. At that magnitude, a double-counting error is a seven-figure mistake. The allocation reconciliation property test is not a formality.

VERIFICATION

```
npx vitest run tests/domain-logic.test.js -t "no double counting"
# Assert across ALL five applicability combinations that
# CORSIA + EU_ETS + UNREGULATED === totalCo2Tonnes exactly
```

RESEARCH

R11 — European Commission general guidance for aircraft operators; EASA EU ETS page.

EFFORT

S

4.12

Offsetting and unit cancellation

ACTION

Record cancelled CORSIA Eligible Emissions Units against the accrued period requirement.

VERIFICATION

```
curl -X POST ../carbon/offsets -d '{"period":"2024-2026","unitsCancelled":120000}'
# Assert cancelled >= accrued requirement for the period, else the
# period cannot be reported as satisfied
```

RESEARCH

R10 — offsetting requires cancelling EEU's equal to the total final requirement for the compliance period.

EFFORT

S

#### 4.13 SAF optimisation value RESEARCH

##### ACTION

Give SAF a computable monetary value so procurement treats it as an investment rather than a compliance box.

```
curl localhost:4000/api/v1/carbon/saf-opportunity
# { safKgPerPeriod: 1_200_000,
#   emissionsAvoidedTonnes: 3792,           // 1.2m × 3.16 / 1000
#   allowanceValueCents: 322_320_00,       // × EUR 85/t
#   recommendedBlendPpm: 200000 }
```

##### WHY THE ZERO FACTOR IS COMMERCIALLY DECISIVE

Under the EU ETS, SAF may be treated with a zero emissions factor. Each kilogram of SAF therefore has a calculable monetary value equal to the allowance price. That turns SAF procurement into a question with a numeric answer, and it belongs in the sourcing module's economics — not in a sustainability report.

##### RESEARCH

*R11 — SAF zero emissions factor; benefit estimated at 0.5 Mt in 2024 rising to 1.7 Mt in 2026.*

##### EFFORT

S

#### 4.14 Regime scope as configuration

##### ACTION

Store thresholds, exemption values, compliance periods and allowance prices as effective-dated reference data.

##### VERIFICATION

```
psql -c "SELECT regime, threshold, effective_from, effective_until
        FROM carbon_thresholds WHERE tenant_id='<x>' ORDER BY effective_from"
# A historical flight must NEVER be re-evaluated against a later threshold.
# Every emission allocation stores the threshold version it used.
```

##### RESEARCH

*R10 — EU ETS scope has been revised repeatedly (geographical restriction extended to start of 2027; scope extending to 5,000 km). A hard-coded constant would be wrong within a year.*

##### EFFORT

S

#### 4.15 Gazette notice generation

##### ACTION

Generate tender, award and standing-list notices in gazette format.

##### VERIFICATION

```
curl localhost:4000/api/v1/sourcing/<id>/gazette-notice -o notice.pdf
# Required fields: reference, description, requirements, quantity,
# unit of measure, closing date and time, contact — per gazette convention
```

##### EFFORT

S

4.16 Wave 4 verification gate

ACTION

Procurement lifecycle, carbon correctness, vendor eligibility. Close only on full green.

```
npm test && npx tsc --noEmit && npm run test:e2e
npm run test:carbon      # reconciliation across all regime combinations
npm run test:sourcing    # full lifecycle including SoD conflicts
```

EFFORT

S

PART IX

# Wave 5 — General Ledger, Close and Reporting

## 5.1 Double-entry journals with a balance trigger

ACTION

Journal header and lines, with a deferred constraint that the document must sum to zero before it can be posted.

```
CREATE OR REPLACE FUNCTION ams_assert_balanced() RETURNS TRIGGER AS $$
DECLARE s BIGINT;
BEGIN
    SELECT SUM(debit_cents) - SUM(credit_cents) INTO s
    FROM journal_lines WHERE journal_id = NEW.journal_id;
    IF s <> 0 THEN
        RAISE EXCEPTION 'Journal % is out of balance by % cents', NEW.journal_id, s;
    END IF; RETURN NULL;
END; $$ LANGUAGE plpgsql;
```

VERIFICATION

```
psql -c "INSERT INTO journal_lines (journal_id, debit_cents, credit_cents) VALUES (1, 100, 99)"
# MUST raise. A one-cent imbalance is exactly the class of error that
# compounds until a trial balance will not foot.
```

EFFORT

M

## 5.2 Period close state machine

ACTION

OPEN → SOFT\_CLOSED → HARD\_CLOSED, with reopen requiring accounting-officer authority and an audit entry.

VERIFICATION

```
curl -X POST .../periods/2026-08/hard-close
# Then: any posting to 2026-08 MUST be rejected by trigger, not by
# application convention. Test it with a direct psql INSERT.
```

EFFORT

S

## 5.3 Suspense ageing and review

ACTION

Every suspense balance older than threshold requires an investigated owner. Unidentified suspense is where misappropriation hides.

VERIFICATION

```
curl localhost:4000/api/v1/suspense/ageing
# { '0-30': 4, '31-90': 11, '91-180': 27, '180+': 3 }
# Anything in 180+ MUST have an owner and an investigation reference
```

EFFORT

S

5.4

Continuous reconciliation against external feeds

ACTION

Match transactions continuously rather than reconciling at month end.

| Feed                        | Matched against  | Finds                                            |
|-----------------------------|------------------|--------------------------------------------------|
| Bank statements             | General ledger   | Unmatched and aged items                         |
| Fuel delivery notes         | Uplifts and burn | Under-billing, misdelivery, uplift mis-recording |
| Lessor statements           | Lease schedule   | Recoverable maintenance offset claims            |
| Payroll                     | Duty reports     | Crew paid but not rostered                       |
| Tariffs and statutory rates | Invoices         | Under-recovery                                   |

THE COMMERCIAL PRIZE

Most carriers under-recover from lessors, fuel suppliers and agents — not through fraud, but because nobody has time to match the detail. Continuous matching typically recovers a fraction of a percent of operating cost, which on a mid-size carrier is a seven-figure line. This is the highest-return feature in the wave.

VERIFICATION

```
curl localhost:4000/api/v1/reconciliation/fuel?period=2026-08
# { matchedKg: 98.7, unmatchedKg: 12400, recoverableCents: ..., agedItems: [...] }
```

RESEARCH

R13 — per-delivery-note capture is what makes fuel reconciliation possible at all.

EFFORT

L

5.5

Route P&L with both levers

ACTION

Shipped in `costing.js`; expose with route and fleet dimensions, and present the commercial-versus-operational distinction explicitly.

VERIFICATION

```
curl 'localhost:4000/api/v1/reports/route-pnl?route=HRE-LUM&period=2026-08'
# { verdict:'MARGINAL', marginPpm: 41_000,
#   levers: { loadFactorGapPpm: -80_000,
#             caskHeadroomMicrocents: -1_200 } }
# Both negative => commercial problem. Cask only => operational problem.
```

EFFORT

S

### 5.6 Fleet performance reporting

ACTION

Utilisation, cycles, hours, check adherence, component removals, fuel efficiency by tail.

VERIFICATION

```
curl localhost:4000/api/v1/reports/fleet-performance
# { tail, blockHoursPerDay, cyclesPerMonth, utilisationPpm,
#   kgPerBlockHourMilli, unplannedRemovals: 2, checkAdherencePpm: 994_000 }
```

EFFORT

S

### 5.7 Cost anomaly detection across the fleet

ACTION

Learn a per-fleet-type, per-season distribution of each cost category and flag divergence.

VERIFICATION

```
curl localhost:4000/api/v1/anomalies?window=30
# { cateringCents: { meanPpm: 9_800_000, zScore: -4.2,
#   flag: 'BELOW_DISTRIBUTION' }, ... }
# A flag RAISES AN INVESTIGATION. It never auto-adjusts a value.
```

RESEARCH

R7 — a distribution-based approach works because industry cost structure is stable enough within a fleet type and season to learn.

EFFORT

M

### 5.8 Executive dashboard

ACTION

Consolidated P&L, fleet performance, route board, cash, carbon exposure, exceptions.

VERIFICATION

```
npm run test:e2e -- executive-dashboard
# Every figure on the dashboard must drill through to source transactions.
# A dashboard number that cannot be traced is not decision-grade.
```

EFFORT

M

5.9

Trial-balance hash publication

RESEARCH

ACTION

Because the audit log is hash-chained and a period is closed, the trial balance becomes a deterministic function of append-only data — so it can be published for external verification.

```
ams verify --period 2026-08 --tenant acme-air
chain verified: 1,842,201 entries, 0 mismatches
trial balance hash: 9f2c1a77e4b0...
anchor: 2026-09-01T00:00:00Z
```

VERIFICATION

```
curl localhost:4000/api/v1/reports/close/2026-08/verify
# Run twice - the hash MUST be identical. Non-determinism here means
# something in the period is mutable, which invalidates the guarantee.
```

EFFORT

S

5.10

Auditor and OAG export

ACTION

Read-only evidence packs with chain verification included.

EFFORT

S

5.11

Hedging ledger

RESEARCH

ACTION

Positions, contracted versus physical volumes, crack spread, mark-to-market, and hedge effectiveness, feeding route cost directly.

EFFORT

M

5.12

Wave 5 verification gate

```
npm test && npx tsc --noEmit && npm run test:e2e
psql -c "INSERT INTO journal_lines (journal_id, debit_cents, credit_cents) VALUES (1,1,0)"
# MUST raise - balance trigger active
```

EFFORT

S

PART X

# Wave 6 — Hardening and Launch

## 6.1 Tenant-isolation fuzzing

ACTION

Three tenants with deliberately similar identifiers, so a query bug surfaces rather than passing by accident.

VERIFICATION

```
npm run test:tenant-isolation
# Every read endpoint fuzzed with other tenants' identifiers across
# path params, query strings, bodies, batch endpoints and filters.
# MUST return 404 - never 403, which discloses existence.
```

EFFORT

M

## 6.2 RLS assertion against raw SQL

ACTION

Execute an unfiltered query directly as the application role and assert only the caller's tenant returns.

```
psql -U ams_app -c "BEGIN; SELECT set_config('app.tenant_id','t-a',true);
                    SELECT count(*) FROM budget_lines WHERE tenant_id='t-b';"
# MUST be 0 - for the OWNER of the table, not just a restricted role
```

EFFORT

S

## 6.3 Concurrency stress beyond the unit test

ACTION

Run the concurrency proof at scale: 500 concurrent authorisations against one line.

```
npx vitest run tests/integration/concurrency.test.js --concurrency=25
# Assert: successes === headroom exactly, and ledger integrity holds.
```

RESEARCH

R12.

EFFORT

S

## 6.4 Load test at 5,000 users

```
k6 run --vus 5000 --duration 15m tests/load/main.js
# Targets: read p95 < 300ms · cost authority p95 < 200ms
#         tender deadline ingestion < 500ms at 5x peak
```

EFFORT

M

## 6.5 Deadline burst simulation

### ACTION

**5,000 bid submissions inside the closing hour, while general API traffic continues.**

### VERIFICATION

```
k6 run tests/load/tender-deadline.js
# ASSERT: bid ingestion stays within budget AND the general API does NOT
# degrade. This is the specific failure mode identified in R14 – a deadline
# surge taking down the whole system.
```

### RESEARCH

*R14 — operational planning systems degrade under load; isolation is the mitigation.*

### EFFORT

**M**

## 6.6 Security suites

```
npm run test:security # IDOR, privilege escalation, injection, audit integrity
npm audit --audit-level=high
trivy image --severity HIGH,CRITICAL ams:latest
```

### EFFORT

**M**

## 6.7 Digital twin — forecast mode

### ACTION

**Extend the costing engine's pure function to evaluate projected inputs rather than actuals.**

```
POST /api/v1/simulate/flight
{ aircraft, route, weightKg, payloadKg, weather, fuelPriceCentsPerKg }
# → { predictedCostCents, uncertaintyPpm, confidence,
#     reserveCompliance: { etopsClass, requiredKg, plannedKg } }
# Uncertainty band derives from the fleet's observed variance distribution.
```

### RESEARCH

*R1 (reserve compliance) · R14 (distinguish a planning estimate from an actual cost — never mix them).*

### EFFORT

**L**

## 6.8 Route comparison

```
POST /api/v1/simulate/compare
{ baseline: {...}, alternative: {...} }
# → { deltaCostCents, deltaEmissionsTonnes, connectionRiskDelta, verdict }
```

### EFFORT

**M**

### 6.9 Backup and PITR verification

ACTION

Monthly automated restore into an isolated environment, verified by ledger and chain integrity.

A BACKUP THAT HAS NEVER BEEN RESTORED IS NOT A BACKUP

The restore drill is a release gate, not a quarterly chore. An unverified backup fails silently and is discovered during an actual incident, which is the worst possible moment to learn that it was empty.

```
npm run dr:restore-drill
# Assert after restore: audit chain verifies, trial balance unchanged,
#     generated available_cents still matches its components
```

EFFORT

M

### 6.10 Observability and alerting

```
curl localhost:4000/api/v1/health
# { status, db, redis, outboxLag, chainHead }
# SEV-1 alerts: audit chain mismatch · negative availability ·
#     outbox depth > 1000 · cross-tenant access attempt
```

EFFORT

S

### 6.11 Runbooks

ACTION

Deploy, rollback, failover, incident response, on-call, and the DR sequence from specification §15.4.

EFFORT

M

### 6.12 Six user manuals

ACTION

Finance, Procurement, Technical, Executive, IT/Admin, Auditor — written for the person doing the job, not for the system.

EFFORT

M

6.13

Compliance attestation

ACTION

Evidence pack mapping every researched obligation to a passing test.

```
npm run attest -- --regulator caaz
# Produces: R1 fuel components → T-x PASS
#           R2 MEL categories + AD subordination → T-x, T-y PASS
#           R6 accountable managers → T-z PASS
#           R10/R11 carbon → T-a PASS
```

RESEARCH

All of R1–R14. This is the artefact that converts research into evidence.

EFFORT

S

6.14

Launch gate

```
npm run gate:launch # consolidates every gate in Part XIV
```

EFFORT

S

## PART XI

# Infrastructure and CI/CD Build-Out

## 11.1 Container and local stack

### 11.1 Single-stage API image — no build step

#### WHY SINGLE-STAGE

Because the codebase is plain JavaScript with no compilation (ADR-001), there is no build artefact to produce and no TypeScript toolchain in the runtime image. This is a direct, concrete benefit of the JavaScript decision worth recording.

```
# backend/Dockerfile
FROM node:20-alpine
WORKDIR /workspace
COPY shared ./shared
COPY backend ./backend
WORKDIR /workspace/backend
RUN npm install --omit=dev
# Run migrations at start, then serve. Nothing to compile.
CMD ["sh", "-c", "node knex-cli.js migrate && node src/server.js"]
```

#### VERIFICATION

```
docker build -f backend/Dockerfile -t ams:latest .
# Assert: NO typescript, NO tsx, NO ts-node in the image
docker run --rm ams:latest sh -c "npm ls --depth=0 | grep -E 'typescript|tsx'" # must be empty
```

#### EFFORT

S

### 11.2 Service orchestration

#### ACTION

PostgreSQL 16, Redis 7, MinIO, API, worker, web, Prometheus, Grafana — already in `docker-compose.yml`.

#### VERIFICATION

```
docker compose up -d
docker compose ps          # all healthy
curl localhost:4000/api/v1/health
```

#### EFFORT

S

### 11.3 Worker scaling on outbox depth, not request rate

#### ACTION

Workers scale on queue depth. API replicas scale on request rate. Conflating them wastes capacity in both directions.

```
docker compose up -d --scale worker=3
# Watch outboxLag fall to zero - it is the scaling signal
```

#### EFFORT

S

## 11.4 CI pipeline

### 11.4 GitHub Actions pipeline

```
jobs:
  test:
    steps:
      - run: npm ci
      - run: npm run lint          # oxlint - zero warnings tolerated
      - run: npx tsc --noEmit      # the ADR-001 gate. NOT advisory.
      - run: npm test              # 118 tests, no services
      - run: npm run test:integration # PostgreSQL service container
      - run: npm run test:security
      - run: npm run test:tenant-isolation
      # ^ THE GATE: concurrency.test.js runs here on every PR.
  build:
    needs: test
    steps:
      - run: docker build -t ams:${{ github.sha }} .
      - run: trivy image --severity HIGH,CRITICAL --exit-code 1 ams:${{ github.sha }}
  deploy:
    needs: build
    # Manual approval gate - four eyes. Production is never
    # deployed by the same push that wrote the code.
    if: github.ref == 'refs/heads/main'
```

#### A GATE THAT IS ADVISORY IS WORSE THAN NO GATE

On first run, `tsc --noEmit` reported 47 defects, including two genuine source bugs the unit tests missed entirely — a `ReferenceError` and an unreachable branch. That is the justification for running it on every pull request rather than occasionally. If it ever begins producing so many warnings that developers start ignoring it, ADR-001 must be revisited rather than the gate muted.

#### EFFORT

M

## 11.5 Observability

11.5

Metrics, logs, traces and business dashboards

| Signal    | Content                                                                                               | Alert                                         |
|-----------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Metrics   | Rate, errors, latency per route · DB pool saturation · outbox depth and lag · cost-authority latency  | p95 > 800ms · outbox lag > 60s                |
| Business  | Tender cycle time · commitment rejection count · budget variance · burn variance · reconciliation age | Commitment rejections > 3× the 30-day average |
| Integrity | Audit chain verification · ledger immutability · RLS assertion · balance trigger                      | Any failure — SEV-1, page immediately         |
| Traces    | OpenTelemetry across API → DB → external. 100% sampling on payment and award paths                    | Trace error rate                              |

EFFORT

M

PART XII

# Test Build Order

Twenty critical tests, in the order they must be written. Order matters: a test written before the code it validates tends to encode a misunderstanding, and the failures during Wave 0 proved that.

## 12.1 Already shipped — 118 tests

| Suite                       | Count | Validates                                                                         | Wave |
|-----------------------------|-------|-----------------------------------------------------------------------------------|------|
| money.test.js               | 17    | Parsing precision, integer discipline, allocation exactness, reconciliation guard | 0    |
| metrics.test.js             | 12    | ASK/RPK, load factor, CASK/RASK, break-even, roll-up correctness                  | 0    |
| costing.test.js             | 23    | Cost build-up, allocation drivers, marginal vs full, route verdict, aggregation   | 0    |
| domain-logic.test.js        | 45    | Airworthiness, fuel, carbon                                                       | 0    |
| audit-costauthority.test.js | 21    | Hash chain, tamper detection, tenant isolation, availability arithmetic           | 0    |

## 12.2 Tests to write next — in order

| #    | Wave | Test                                                  | Pass condition                                                          | Research |
|------|------|-------------------------------------------------------|-------------------------------------------------------------------------|----------|
| T-01 | 2    | Over-commitment                                       | 422 with correct available balance and shortfall; no ledger row written | —        |
| T-02 | 2    | Concurrency: 50 parallel, headroom 10                 | Exactly 10 succeed; 40 denied; ledger sums to headroom; available ≥ 0   | R12      |
| T-03 | 2    | Payment does not restore availability                 | Available unchanged; committed down; expended up                        | —        |
| T-04 | 2    | Cancelled requisition restores availability           | Available restored; RELEASE ledger row exists                           | —        |
| T-05 | 2    | Negative availability is not clamped                  | Throws AVAILABILITY_NEGATIVE                                            | —        |
| T-06 | 2    | Payment without receipt                               | Blocked; exception recorded                                             | —        |
| T-07 | 1    | Cross-tenant read via path parameter                  | 404, no count, no existence disclosure                                  | —        |
| T-08 | 1    | Cross-tenant read via crafted filter                  | Zero rows; filter not overridable by query string                       | —        |
| T-09 | 0    | Audit tampering: edit · delete · reorder · swap actor | All four detected                                                       | —        |
| T-10 | 0    | One tenant's audit corruption                         | t-1 fails, t-2 still verifies                                           | —        |
| T-11 | 4    | Sub-threshold sourcing attempt                        | Method overridden; direct award requires recorded exception             | —        |

| #    | Wave | Test                                           | Pass condition                                            | Research |
|------|------|------------------------------------------------|-----------------------------------------------------------|----------|
| T-12 | 4    | Opening without a constituted committee        | Refused; no register generated                            | R6       |
| T-13 | 4    | Vendor bids on its own sourcing event          | Submission rejected                                       | —        |
| T-14 | 0    | Allocation sums to pool across all six drivers | Zero leakage on a real US\$486,213.88 pool                | R9       |
| T-15 | 0    | Money arithmetic precision                     | One cent × 1,000,000 = exactly 1,000,000                  | —        |
| T-16 | 0    | Carbon allocation reconstructs the total       | Exact across all five applicability combinations          | R10/R11  |
| T-17 | 0    | AD applied does not satisfy an SB              | <code>requiresMandatoryInstruction</code> differs by kind | R3       |
| T-18 | 0    | LLP controlling counter when calendar binds    | Correct counter despite generous cycle life               | R3       |
| T-19 | 0    | Landing reserve below regulatory minimum       | <code>reservesMeetRegulatoryMinimum === false</code>      | R1       |
| T-20 | 0    | Roll-up recomputes rather than averages ratios | Capacity-weighted result differs from naive mean          | R8       |
| T-21 | 3    | MEL deferral cannot cure an AD                 | AD still OVERDUE; deferral reported as illegal            | R2       |
| T-22 | 3    | CAT B excludes the day of discovery            | Monday 09:00 → Thursday 23:59:59, not Thursday 09:00      | R2       |
| T-23 | 4    | CORSIA threshold and new-entrant exemption     | Below 10,000 t not obligated; 3-year exemption honoured   | R10      |
| T-24 | 5    | Trial-balance hash is reproducible             | Two verifications return the identical hash               | —        |
| T-25 | 5    | Journal balance trigger                        | One-cent imbalance rejected at database level             | —        |
| T-26 | 6    | Restore-from-PITR integrity                    | Chain verifies; trial balance unchanged after restore     | —        |

#### A LESSON FROM WAVE 0 WORTH CARRYING FORWARD

Three of my own test assertions during Wave 0 were wrong. One wrote `cents('1,850,000')` believing it was \$18,500 — it is \$1.85M. Another asserted a tankering outcome whose premise was economically backwards, which exposed a genuine design flaw in the tankering model. Tests that encode a misunderstanding fail loudly, which is far cheaper than tests that encode a misunderstanding and pass. Write the test before the code wherever the requirement comes from research rather than from implementation.

PART XIII

# Data Onboarding and Reference Data

## 13.1 Seed the cost taxonomy from research, not invention

```
npm run seed -- cost-taxonomy
# 22 categories across 3 buckets, seeded from IATA-aligned industry
# structure. Every seed value carries a source reference column.
```

VERIFICATION

```
psql -c "SELECT bucket, count(*), string_agg(code, ' ' ORDER BY code)
        FROM cost_categories GROUP BY bucket"
# FLIGHT_ATTRIBUTABLE | 11 | fuel, crew_flight, crew_cabin, ...
# FLEET_FIXED         | 5 | lease_aircraft, insurance_aircraft, ...
# PERIOD              | 6 | sg_a, g_a, finance_cost, tax, ...
```

RESEARCH

R7, R9. The seed must be traceable to an industry source; an invented category would be excluded from every third-party benchmark.

EFFORT

S

## 13.2 Seed carbon thresholds and compliance periods

```
npm run seed -- carbon-thresholds
# CORSIA: 10,000 t threshold · 5,700 kg MTOM · 3-year new-entrant
#          0.1% of 2020 baseline · compliance periods 2021-23 ... 2033-35
# EU ETS: 243 flights / 10,000 t thresholds · 5,000 km scope · CORSIA deduction
# Effective-dated. A regulatory change is a DATA UPDATE, not a deployment.
```

VERIFICATION

```
node -e "
import('./shared/src/carbon.js').then(c=>{
  console.log(c.corsiaCompliancePeriod(2024)); // '2024-2026'
  console.log(c.corsiaCompliancePeriod(2027)); // '2027-2029'
  try { c.corsiaCompliancePeriod(2040); }      // must throw RangeError
  catch(e) { console.log('correctly refused'); }
});"
```

RESEARCH

R10, R11.

EFFORT

S

### 13.3 Seed the regulatory post roster

```
npm run seed -- accountable-managers
# The six mandated posts from R6, created UNFILLED so the platform
# surfaces the gap as a compliance exception on day one.
# ceo · flight_ops · aircraft · safety_officer
# quality_manager · security_manager
```

#### VERIFICATION

```
curl localhost:4000/api/v1/compliance/accountable-managers
# { compliant: false, unfilled: [ all six ],
#   note: 'An operator cannot hold or renew its AOC without evidencing
#         these posts (R6).' }
```

#### RESEARCH

R6.

#### EFFORT

XS

### 13.4 Fleet import with AD applicability pre-computation

#### ACTION

Import the aircraft register, then evaluate every applicable AD against every aircraft and write the compliance board in one pass.

```
npm run import -- fleet --file aircraft.csv
# Required columns: registration, msn, type_code, mtom_kg,
#                  seats_installed, lease_type, first_flight_date
npm run evaluate -- adsb --tenant acme-air
# Produces one adsb_compliance row per (aircraft, applicable AD)
# including NON-applicable determinations, so an engineer can see
# that a directive was considered and ruled out – not merely missed.
```

#### VERIFICATION

```
psql -c "SELECT status, count(*) FROM adsb_compliance GROUP BY status"
# COMPLIANT | 412 · DUE_SOON | 38 · OVERDUE | 3 · NOT_APPLICABLE | 1_847
# NOT_APPLICABLE MUST be recorded, not omitted – an unrecorded
# non-applicability is indistinguishable from a missed directive.
```

#### RESEARCH

R3 — applicability depends on model, serial range and embodied modifications, so a naive type-only match will produce false positives.

#### EFFORT

M

### 13.5 Opening balance migration

#### ACTION

Migrate existing ledger balances as opening entries with full provenance, so the pre-AMS period is auditable rather than a lump sum.

```
npm run import -- opening-balances --file opening.csv --dry-run
# DRY RUN FIRST. The report must show every line's destination cost
# centre, fund, category and period BEFORE anything is written.
# Then assert the generated available_cents per line is non-negative.
```

#### VERIFICATION

```
psql -c "SELECT count(*) FROM budget_lines WHERE available_cents < 0"
# MUST be 0 - if it is not, the opening balances were wrong, and the
# CHECK constraint caught it before any transaction was possible
```

#### EFFORT

M

### 13.6 First-year parallel run

#### ACTION

Run AMS alongside the existing process for one full financial year, reconciling monthly. Do not cut over on a go-live date.

#### VERIFICATION

```
curl localhost:4000/api/v1/reports/parallel-run?month=2026-04
# { legacyCents, amsCents, varianceCents, variancePpm,
#   unexplainedItems: [] }
# Target: variance → 0 with every difference explained, not just small.
```

#### WHY A PARALLEL RUN RATHER THAN A CUTOVER

A cutover makes the first month unreconcilable by definition, and the temptation is to write it off. A parallel run makes every difference an item to be explained, which is how the cost model gets corrected against reality before anyone depends on it.

#### EFFORT

L

PART XIV

# Consolidated Verification Gates

## 14.1 Per-wave gate

| Wave | Must pass before closing                                                                                                                                                                                                                                                                                         |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| W0   | All 118 unit tests · <code>tsc --noEmit</code> clean · migrations apply and roll back · audit trigger raises on UPDATE and DELETE · <code>available_cents</code> is GENERATED                                                                                                                                    |
| W1   | Refresh replay revokes the session · cross-tenant read returns 404 with no disclosure · self-approval refused with no ledger row · delegation cannot exceed the delegator's ceiling · chain verifies · all 16 departments seeded with accountable roles                                                          |
| W2   | Concurrency proof: 50 parallel, headroom 10 ⇒ exactly 10 succeed · over-commitment 422 with shortfall · payment does not restore availability · cancellation restores it · journal balance trigger active · cost authority p95 < 200ms                                                                           |
| W3   | MEL deferral cannot cure an AD (T-21) · CAT B excludes the day of discovery (T-22) · AD applicability by serial range and modifications · NOT_APPLICABLE determinations recorded · dispatch gate blocks on overdue · LLP controlling counter correct across all three counters                                   |
| W4   | Carbon allocation reconciles across all regime combinations · CORSIA threshold and exemption correct · SAF value computed · sourcing cannot publish without a plan line · opening requires constituted committee · award blocked against an ineligible vendor · award requires post-qualification and preference |
| W5   | Journal balance trigger enforced at database level · trial-balance hash reproducible across two runs · close blocks without a cash forecast · suspense ageing surfaces 180+ items · route P&L levers both computed                                                                                               |
| W6   | Zero cross-tenant leakage across the full adversarial suite · 5,000-user load targets met · deadline burst does not degrade the general API · PITR restore verified · chain verifies after restore · compliance attestation complete                                                                             |

## 14.2 Launch criteria

| #    | Criterion                                                         | Evidence                                                       |
|------|-------------------------------------------------------------------|----------------------------------------------------------------|
| A-1  | A full financial year processed from budget to audited statements | Demo-year run, all reports produced                            |
| A-2  | Over-commitment impossible under concurrent load                  | T-02 passing repeatedly, including with the lock removed       |
| A-3  | Zero cross-tenant leakage                                         | Adversarial suite green; RLS assertion from a raw psql session |
| A-4  | Every financial figure traceable to source transactions           | Auditor drill-down demonstration                               |
| A-5  | Any reported number reproducible from append-only data            | T-24 trial-balance hash, twice                                 |
| A-6  | CASK reconciles to a third-party benchmark                        | Reconciliation to an IATA-aligned external source              |
| A-7  | Carbon compliance bill reproducible per flight                    | T-16 plus regulator-facing extract                             |
| A-8  | Audit chain verifies and detects tampering                        | T-09, T-10                                                     |
| A-9  | Restore-from-PITR within the recovery objective                   | T-26 drill report                                              |
| A-10 | A tender can be run end to end without training                   | First-time-user walkthrough                                    |
| A-11 | Every researched obligation maps to a passing test                | Compliance attestation across R1–R14                           |

PART XV

# Research Bibliography

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Fourteen sources. Each states what it constrained in the build, so a future maintainer can tell whether a source is still load-bearing.

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1<br>Fuel components      | <p>UK CAA SPO.OP.131 / SPA.NCO.131 — <i>Fuel and oil supply</i>. Corroborated by EASA fuel-planning guidance and ICAO <i>Flight Planning and Fuel Management</i> seminar material (2021). <a href="https://regulatorylibrary.caa.co.uk">regulatorylibrary.caa.co.uk</a> · <a href="https://easa.europa.eu">easa.europa.eu</a> · <a href="https://icao.int">icao.int</a></p> <p><b>Constrains:</b> steps 0.8, 3.6, 3.7. Nine named fuel components; 30-minute final reserve for turbine aeroplanes at 1,500 ft above the alternate, ISA.</p>                                                            |
| R2<br>MEL structure        | <p>EASA CS-GEN-MMEL Issue 2, Annex II to ED Decision 2020/012/R. Corroborated by UK CAA <i>MEL Compliance Document</i>, Edition 8, February 2024. <a href="https://easa.europa.eu">easa.europa.eu</a> · <a href="https://caa.co.uk">caa.co.uk</a></p> <p><b>Constrains:</b> steps 3.9, 3.10, 3.11; tests T-21, T-22. Category A/B/C/D rectification intervals; day of discovery excluded; five-column MMEL item table; (M) and (O) symbols; "The MEL cannot deviate from Airworthiness Directives".</p>                                                                                                |
| R3<br>AD structure         | <p>US 14 CFR Part 39 as published in the Federal Register — 79 FR 19848 (April 2014) and 87 FR 67354 (November 2022). Live EASA directives: AD 2026-0125-E (Emergency), AD 2023-0167R1, AD 2024-0213. <a href="https://govinfo.gov">govinfo.gov</a> · <a href="https://ad.easa.europa.eu">ad.easa.europa.eu</a> · <a href="https://caat.or.th">caat.or.th</a></p> <p><b>Constrains:</b> steps 1.3, 3.8, 3.12, 3.13; test T-17. Modification-based applicability, multiple compliance thresholds, repetitive intervals, ATA chapter, 5% non-cumulative tolerance, terminating action, supersession.</p> |
| R4<br>ICAO Annex 6         | <p>ICAO Annex 6 — Operation of Aircraft, Parts I–IV, 11th edition and amendments (including Part IV RPAS, adopted April 2024). <a href="https://icao.int">icao.int</a></p> <p><b>Constrains:</b> steps 0.9, 1.6, 3.12. AOC template and Operations Specifications (Appendix 6, Attachment D); Operations Manual (Attachment G); MEL (Attachment E); EDTO (Attachment C); RVSM (Appendix 4).</p>                                                                                                                                                                                                        |
| R5<br>Personnel licensing  | <p>ICAO Annex 1 — Personnel Licensing. Civil Aviation Authority of Zimbabwe — personnel licensing and training-organisation approval per ICAO and national requirements, alongside AFCAC, Immigration, ZIMRA-Customs and Meteorological Services. <a href="https://icao.int">icao.int</a> · <a href="https://caaz.co.zw">caaz.co.zw</a></p> <p><b>Constrains:</b> steps 1.7, 4.2. Licence, medical and recurrent training are three separate expiries, not one validity date.</p>                                                                                                                      |
| R6<br>Accountable managers | <p>South African Civil Aviation Regulations 127.06.2 (Commercial Helicopter Operations) — appointed personnel enumerated. Corroborated by the Mauritius Department of Civil Aviation <i>Guidance for obtaining an Air Operator Certificate</i>, Issue 1 Rev 0, aligned to EU 2018/1139 and ICAO Doc 8335. <a href="https://acts.co.za">acts.co.za</a> · <a href="https://civil-aviation.govmu.org">civil-aviation.govmu.org</a></p> <p><b>Constrains:</b> steps 1.2, 1.5, 1.6, 13.3. Six mandated posts, approved not merely filled. This is the origin of the 16-department registry.</p>             |
| R7<br>Cost structure       | <p>US BTS Table 4 — Quarterly US Scheduled Passenger Airlines Revenue, Expenses and Profits (25 airlines, 3Q 2016). IATA cost taxonomy. IATA Revenue Accounting Manual B16. ATA Spec 8. <a href="https://bts.gov">bts.gov</a> · <a href="https://iata.org">iata.org</a></p> <p><b>Constrains:</b> steps 0.2, 2.4, 5.8. Industry mix: labour 34.31%, fuel 16.41%, rentals 5.99%, D&amp;A 5.70%, landing fees 2.10%, maintenance materials 1.64%. Corrects the assumption that fuel is the largest cost — hence crew is flight-attributable.</p>                                                         |
| R8<br>Traffic metrics      | <p>IATA — <i>Understanding air traffic metrics</i> (Monthly Air Traffic Statistics, 93% of world scheduled traffic). Corroborated by the MIT / PubMed Central study <i>Operating costs and capacity in the airline industry</i>. <a href="https://iata.org">iata.org</a> · <a href="https://pmc.ncbi.nlm.nih.gov">pmc.ncbi.nlm.nih.gov</a></p> <p><b>Constrains:</b> steps 0.3, 2.5, 5.9; test T-20. ASK, RPK, PLF, CTk, ACTk, CLF, ATK, FRT. ASK is <i>available</i> capacity — sold, boarded and no-show counts must be distinct.</p>                                                                |
| R9<br>Cost allocation      | <p>MIT / PubMed Central, <i>Operating costs and capacity in the airline industry</i> — activity-based costing applied to aviation, separating direct variable, direct fixed and period costs. <a href="https://pmc.ncbi.nlm.nih.gov">pmc.ncbi.nlm.nih.gov</a></p> <p><b>Constrains:</b> steps 0.2, 2.6, 2.7; test T-14. Three-bucket taxonomy, and passenger/cargo separation by causality of cost occurrence before unit cost is computed.</p>                                                                                                                                                        |
| R10<br>CORSIA              | <p>ICAO Assembly Resolution A41-22; CORSIA FAQ (December 2022); Annex 16 Volume IV; Environmental Technical Manual Doc 9501 Vol IV; IATA CORSIA programme page. <a href="https://icao.int">icao.int</a> · <a href="https://iata.org">iata.org</a></p> <p><b>Constrains:</b> steps 0.7, 4.10, 4.12, 13.2; test T-23. 10,000 t CO<sub>2</sub> threshold · 5,700 kg MTOM exemption · 3-year new-entrant exemption (or 0.1% of 2020 baseline) · compliance periods 2021-23 ... 2033-35 · 2019-2020 baseline, 85% for 2024-2035.</p>                                                                        |
| R11<br>EU ETS              | <p>European Commission, <i>General guidance for Aircraft Operators — EU ETS</i>. EASA EU Emissions Trading System page. European Commission <i>Questions and answers on the ETS review</i>, 17 July 2026. <a href="https://climate.ec.europa.eu">climate.ec.europa.eu</a> · <a href="https://easa.europa.eu">easa.europa.eu</a> · <a href="https://ec.europa.eu">ec.europa.eu</a></p> <p><b>Constrains:</b> steps 0.7, 4.11, 4.13, 4.14; test T-16. In the ETS since 2012 · &gt;EUR 80/tonne in 2022-23 · EUR</p>                                                                                      |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 1.8bn (2022) and EUR 2.1bn (2023) sector cost · 4-6% of operating cost 2024-26 · 5,000 km scope extension · CORSIA deduction · non-CO <sub>2</sub> MRV from 1 Jan 2025 · SAF zero emissions factor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| R12<br>Row locking       | PostgreSQL documentation §13.2 <i>Transaction Isolation</i> (semantics unchanged since 8.1). Corroborated by CockroachDB's <i>FOR UPDATE</i> and <i>FOR SHARE</i> documentation on the same semantics. <a href="https://postgresql.org/docs/current/transaction-iso.html">postgresql.org/docs/current/transaction-iso.html</a><br><i>Constrains:</i> steps 2.4, 2.5, 6.3; test T-02 — the most important test in the project. Read Committed is the default; on <i>FOR UPDATE</i> a blocked transaction receives " <i>the updated version of the row</i> ". Also why <i>SERIALIZABLE</i> must not be used: it surfaces serialization errors into a business workflow.                                                                                                                                                                                                                         |
| R13<br>Into-plane fuel   | IATA — <i>Standard Into-Plane Fueling Service Levels and Safety</i> (v1.00), referencing ICAO Annexes 6 and 14 provisions on fuelling with passengers on board. <a href="https://www.iata.org">iata.org</a><br><i>Constrains:</i> steps 3.5, 3.7, 4.5. Price per kg captured per delivery note, per station, per date — never a period average, because into-plane cost drives network decisions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| R14<br>Incumbent systems | Vendor product documentation surveyed: Lufthansa Systems NetLine/Sched · NetLine/Ops · NetLine/Crew; PDC Airline Suite (FlightTime · FlightOps · FlightCrew · GroundCost · FlightBag); AIMS Airline Software Commercial Planning and Crew Management; SOMA Software Flight Operations; SmartOPS aviation modules for Odoo. <a href="https://lhsystems.com">lhsystems.com</a> · <a href="https://pdc.com">pdc.com</a> · <a href="https://aimsairlinesoftware.com">aimsairlinesoftware.com</a> · <a href="https://somasoftware.com">somasoftware.com</a> · <a href="https://smartops.aero">smartops.aero</a><br><i>Constrains:</i> steps 2.6, 5.9, 6.7, 6.8. Incumbents compute <i>estimates</i> in the planning workspace (Tactical PEM, FlightCost & Revenue Analysis). AMS computes <i>actuals</i> from primary records. Conflating the two is how two numbers arise with no reconciliation. |

Background sources — the government specification AMS descends from

|      |                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PFMS | S.I. 144 of 2019 — Public Finance Management (Treasury Instructions), 2019. s.154/177 (IMF GFSM chart of accounts); s.178 (SAP EEC6 EHP7 prescribed; independent systems must be SAP-compatible); s.179 (fiscal year 1 Jan – 31 Dec); s.180 (USD functional currency); s.101–110 (assets); s.195 (offline units and weekly commitment capture). <a href="https://www.veritaszim.net">veritaszim.net</a> |
| e-GP | Procurement Regulatory Authority of Zimbabwe — Electronic Government Procurement System. Supplier category codes ( <i>GT002</i> , <i>ST002</i> , <i>GS002</i> , <i>GN003</i> ), tender detail structure, UNSPSC line items, bid security and establishment amounts, APP panels. <a href="https://egp.praz.org.zw">egp.praz.org.zw</a>                                                                   |
| PEFA | PEFA Assessment 2017 — Government of Zimbabwe with the World Bank, under the PFM Enhancement Project (ZIMREF). Independent benchmark used to identify the weaknesses that shaped the control discipline inherited into AMS. <a href="https://pefa.org">pefa.org</a>                                                                                                                                     |

WHAT THE RESEARCH CHANGED, IN ONE PARAGRAPH

Fourteen sources produced three mandatory design corrections (Part III): a MEL deferral must never be permitted to cure an AD, because the two are legally subordinate; ADs must carry ATA chapter, because that is how engineers actually search; and MEL rectification deadlines must exclude the day of discovery, because the shipped duration arithmetic was an off-by-one on every Category B item. It also corrected one assumption carried in the specification — that fuel is the largest airline cost, when the industry aggregate shows labour exceeding it — which is why crew cost is modelled as flight-attributable rather than a period overhead. And it supplied the PostgreSQL citation that proves the cost-authority control works rather than merely being plausible.

End of document · AMS — Step-by-Step Implementation Plan v1.0 · 30 September 2026

7 waves · 96 numbered steps · 26 critical tests · 14 research sources · 3 design corrections

Companion: [AMS-Specification.pdf](#)